

# The Functions in the Package `gamlss.prepdata`

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## Introduction

This booklet introduces the `gamlss.prepdata` package and its functionality. It aims to describe the available functions and how they can be used.

The latest versions of the packages `gamlss`, `gamlss2` and `gamlss.prepdata` are shown below:

```
rm(list=ls())  
library(gamlss)  
library(gamlss2)  
library(ggplot2)  
library(gamlss.ggplots)  
library(gamlss.prepdata)  
library("dplyr")  
packageVersion("gamlss")
```

```
[1] '5.4.23'
```

```
packageVersion("gamlss2")
```

```
[1] '0.1.0'
```

```
packageVersion("gamlss.prepdata")
```

```
[1] '0.1.12'
```

## The package: `gamlss.prepdata`

The `gamlss.prepdata` package originated from the `gamlss.ggplots` package. As `gamlss.ggplots` became too large, for easy maintenance, it was split into two separate packages, and `gamlss.prepdata` was created.

Since `gamlss.prepdata` is still at an experimental stage, some of functions are remained hidden to allow time for thorough checking and validation. These hidden functions can still be accessed using the triple colon notation, for example: `gamlss.prepdata:::`.

The functions available in `gamlss.prepdata` are intended for pre-fitting — that is, to be used before applying the `gamlss()` or `gamlss2()` fitting functions. The available functions can be grouped into the following categories:

### Information functions

These functions provide information about:

- the size of the dataset;
- the extent of missing values;
- the structure of the dataset;
- whether the class of variables is appropriate for analysis.

### Plotting functions

These functions allow plotting;

- individual variables and
- pairwise relationships of variables

### Features functions

These functions can assist in:

- detecting outliers;
- applying transformations to variables and
- scaling variables.

## Data Partition functions

Functions that facilitate partitioning the data to improve inference and avoid overfitting during model selection.

## Distributional Regression

The aim of this vignette is to demonstrate how to manipulate and prepare data before applying a distributional regression analysis.

The general form a distributional regression model can be written as;

$$\begin{aligned} \mathbf{y} &\overset{ind}{\sim} \mathcal{D}(\theta_1, \dots, \theta_k) \\ g_1(\theta_1) &= \mathcal{ML}_1(\mathbf{x}_{11}, \mathbf{x}_{21}, \dots, \mathbf{x}_{p1}) \\ &\dots = \dots \\ g_k(\theta_k) &= \mathcal{ML}_k(\mathbf{x}_{1k}, \mathbf{x}_{2k}, \dots, \mathbf{x}_{pk}). \end{aligned}$$

where we assume that the response variable  $y_i$  for  $i = 1, \dots, n$ , is independently distributed having a distribution  $\mathcal{D}(\theta_1, \dots, \theta_k)$  with  $k$  parameters and where all parameters could be effected by the explanatory variables  $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p$ . The  $\mathcal{ML}$  represents **any** regression type machine learning algorithm i.e. LASSO, Neural networks etc.

When only **additive smoothing** terms are used in the fitting the model can be written as;

$$\begin{aligned} \mathbf{y} &\overset{ind}{\sim} \mathcal{D}(\theta_1, \dots, \theta_k) \\ g_1(\theta_1) &= b_{01} + s_1(\mathbf{x}_{11}) + \dots, + s_p(\mathbf{x}_{p1}) \\ &\dots = \dots \\ g_k(\theta_k) &= b_{0k} + s_1(\mathbf{x}_{1k}) + \dots, + s_p(\mathbf{x}_{pk}). \end{aligned}$$

which is the GAMLSS model introduced by Rigby and Stasinopoulos (2005).

There are three books on GAMLSS, D. M. Stasinopoulos et al. (2017), Rigby et al. (2019) and M. D. Stasinopoulos et al. (2024) and several 'GAMLSS lecture materials, available from GitHub, [https://mstasinopoulos.github.io/Porto\\_short\\_course/](https://mstasinopoulos.github.io/Porto_short_course/) and <https://mstasinopoulos.github.io/ShortCourse/>. The latest **R** packages related to GAMLSS can be found in <https://gamlss-dev.r-universe.dev/builds>.

The aim of this package is to prepare data and extract useful information that can be utilized during the modeling stage.

## Information functions

The functions for obtaining information from a dataset are described in this section and summarized in Table 1. These functions can provide:

- General information about the dataset such as the dimensions (number of rows and columns) and the percentage of omitted (missing) observations
- Information about the variables in the dataset
- Information about the observations in the dataset
- Changing the variables into suitable **R** classes for further analysis.

The functions which use a `data.frame` as their first argument are starting from `data_` while ones variable whixg take only a variable start fro, `y_`.

Here is a table of the information functions;

Table 1: A summary table of the functions to obtain information

| Functions                         | Usage                                                                                             |
|-----------------------------------|---------------------------------------------------------------------------------------------------|
| <code>data_dim()</code>           | Returns the number of rows and columns, and show the percentage of omitted (missing) observations |
| <code>data_names()</code>         | Lists the names of the variables in the dataset                                                   |
| <code>data_rename()</code>        | Allows renaming of one or more variables in the dataset                                           |
| <code>data_shorter_names()</code> | Allows shortening the names of one or more variables in the dataset                               |
| <code>data_distinct()</code>      | Displays the number of distinct values for each variable in the dataset                           |
| <code>data_na_vars()</code>       | Displays the count of NA (missing) values for each variable in the dataset                        |
| <code>data_na_obs()</code>        | Displays which observation are missing for each variable                                          |
| <code>data_omit()</code>          | Removes all rows (observations) that contain NA's (missing values)                                |
| <code>data_str()</code>           | Displays the class for each variable (e.g., numeric, factor, etc.) along with additional detail   |
| <code>data_cha2fac()</code>       | Converts all character variables in the dataset to factors                                        |

| Functions                           | Usage                                                                                                                   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <code>data_few2fac()</code>         | Converts variables with a small number of distinct values (e.g., binary or categorical) to factors                      |
| <code>data_int2num()</code>         | Converts integer variables with several distinct values to numeric                                                      |
| <code>data_fac2num()</code>         | Converts factors or characters to numeric                                                                               |
| <code>data_rm()</code>              | Removes one or more variables (columns) from the dataset                                                                |
| <code>data_rmNAvars()</code>        | Removes all variables with NA's as their only value.                                                                    |
| <code>data_rm1val()</code>          | Removes all variables/factors with a single value/level                                                                 |
| <code>data_select()</code>          | Allows selection of one or more variables (columns)                                                                     |
| <code>data_exclude_class()</code>   | Removes all variables of a specified class (e.g., factor, numeric) from the dataset                                     |
| <code>data_only_continuous()</code> | Retains only the numeric variables (excluding factors or other variable types)                                          |
| <code>data_factor()</code>          | For all factors in the data the first level becomes the level with <b>lower</b> of <b>higher</b> number of observations |

Next we demonstrate simple use of the functions.

#### `data_dim()`

This function provides detailed information about the dimensions of a `data.frame`. It is similar to the R function `dim()`, but with additional details. The output is the original data frame, allowing it to be used in a series of piping commands.

```
rent99 |> data_dim()
```

```
*****
*****
the R class of the data is: data.frame
the dimensions of the data are: 3082 by 9
number of observations with missing values: 0
% of NA's in the data: 0 %
*****
*****
```

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### `data_names()`

This function provides the names of the variables in the `data.frame`, similar to the R function `names()`.

```
rent99 |> data_names()
```

```
*****
*****
the names of variables
[1] "rent"      "rentsqm"   "area"      "yearc"     "location"  "bath"      "kitchen"
[8] "cheating"  "district"
*****
*****
```

The output of the function is the original data frame.

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### `data_rename()`

Renames one or more variables (columns) in the `data.frame` using the function `data_rename()`;

```
da<- rent99 |> data_rename(olddname="rent", newname="R")
head(da)
```

|   | R        | rentsqm   | area | yearc | location | bath | kitchen | cheating | district |
|---|----------|-----------|------|-------|----------|------|---------|----------|----------|
| 1 | 109.9487 | 4.228797  | 26   | 1918  | 2        | 0    | 0       | 0        | 916      |
| 2 | 243.2820 | 8.688646  | 28   | 1918  | 2        | 0    | 0       | 1        | 813      |
| 3 | 261.6410 | 8.721369  | 30   | 1918  | 1        | 0    | 0       | 1        | 611      |
| 4 | 106.4103 | 3.547009  | 30   | 1918  | 2        | 0    | 0       | 0        | 2025     |
| 5 | 133.3846 | 4.446154  | 30   | 1918  | 2        | 0    | 0       | 1        | 561      |
| 6 | 339.0256 | 11.300851 | 30   | 1918  | 2        | 0    | 0       | 1        | 541      |

The output of the function is the original data frame with new names.

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## `data_shorter_names()`

If the variables in the dataset have very long names, they can be difficult to handle in formulae during modelling. The function `data_shorter_names()` abbreviates the names of the explanatory variables, making them easier to use in formulas.

```
rent99 |> data_shorter_names()
```

```
*****
*****
the names of variables
[1] "rent" "rents" "area" "yearc" "locat" "bath" "kitch" "cheat" "distr"
*****
*****
```

If no long variable names exist in the dataset, the function `data_shorter_names()` does nothing. However, when applicable, the function abbreviates long names and returns the original data frame with the new shortened names.

### Warning

Note that there is a risk when using a small value for the `max` option, as it may result in identical names for different variables. This could lead to confusion or errors in the modelling process. It is important to carefully choose an appropriate value for `max` to avoid this issue.

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## `data_distinct()`

The distinct values for each variable are shown below.

```
rent99 |> data_distinct()
```

| rent | rentsqm | area | yearc | location | bath | kitchen | cheating |
|------|---------|------|-------|----------|------|---------|----------|
| 2723 | 3053    | 132  | 68    | 3        | 2    | 2       | 2        |

district  
336

The output of the function is the original data frame.

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### `data_na_vars()`

This function provides information about which variables have missing observations and how many missing values there are for each variable;

```
rent99 |> data_na_vars()
```

```
      rent rentsqm   area   yearc location   bath kitchen cheating  
      0         0       0       0         0       0         0         0  
district  
      0
```

The output of the function is the original data frame nothing is changing.

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### `data_na_obs()`

This function provides information about which variables have missing observations and how many missing values there are for each variable;

```
rent99 |> data_na_obs()
```

```
$rent  
integer(0)
```

```
$rentsqm  
integer(0)
```

```
$area  
integer(0)
```

```
$yearc  
integer(0)
```

```
$location  
integer(0)
```

```
$bath  
integer(0)
```

```
$kitchen
```



```
integer(0)
```

```
$cheating  
integer(0)
```

```
$district  
integer(0)
```

The output of the function is the original data frame nothing is changing.

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```
data_omit()
```

The function `data_omit()` omits all observations with missing values.

```
rent99 |> data_omit()
```

```
*****  
*****  
the R class of the data is: data.frame  
the dimensions of the data before omission are: 3082 x 9  
the dimensions of the data saved after omission are: 3082 x 9  
the number of observations omitted: 0  
*****  
*****
```

The output of the function is a new data frame withll NA's omitted.

#### Warning

It is important to select the relevant variables for the analysis before using the `data_omit()` function, as some unwanted variables may contain many missing values that that could lead to unnecessary omissions.

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`data_str()`

The function `data_str()` (similar to the **R** function `str()`) provides information about the types of variable exist in the data.

```
rent99 |> data_str()
```

```
*****
*****
the structure of the data
'data.frame':  3082 obs. of  9 variables:
 $ rent      : num  110 243 262 106 133 ...
 $ rentsqm   : num   4.23 8.69 8.72 3.55 4.45 ...
 $ area      : int   26 28 30 30 30 30 31 31 32 33 ...
 $ yearc     : num  1918 1918 1918 1918 1918 ...
 $ location: Factor w/ 3 levels "1","2","3": 2 2 1 2 2 2 1 1 1 2 ...
 $ bath      : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ kitchen   : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 2 1 1 ...
 $ cheating  : Factor w/ 2 levels "0","1": 1 2 2 1 2 2 1 2 1 1 ...
 $ district: int   916 813 611 2025 561 541 822 1713 1812 152 ...
*****
*****
table of the class of variabes

factor integer numeric
      4         2         3
*****
*****
distinct values in variables
      rent  rentsqm    area  yearc location    bath  kitchen cheating
      2723    3053     132     68         3         2         2         2
district
      336
consider to make those characters vectors into factors:
location bath kitchen cheating
*****
*****
```

The output of the function is the original data frame.

The next set of functions manipulate variables withing `data.frames`.

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### `data_cha2fac()`

Often, variables in datasets are read as character vectors, but for analysis, they may need to be treated as factors. This function transforms any character vector (with a relatively small number of distinct values) into a factor.

```
rent99 |> data_cha2fac() -> da
```

```
*****
not character vector was found
```

Since no character were found nothing have changed. The output of the function is a new data frame.

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### `data_few2fac()`

There are occasions when some variables have very few distinct observations, and it may be better to treat them as factors. The function `data_few2fac()` converts (numeric) vectors with a small number of distinct values into factors.

```
rent99 |> data_few2fac() -> da
```

```
*****
      rent  rentsqm    area  yearc location    bath  kitchen cheating
      2723    3053    132    68         3        2        2        2
district
      336
*****
4 vectors with fewer number of values than 5 were transformed to factors
*****
*****
```

```
str(da)
```

```
'data.frame':  3082 obs. of  9 variables:
 $ rent      : num  110 243 262 106 133 ...
 $ rentsqm   : num  4.23 8.69 8.72 3.55 4.45 ...
 $ area      : int  26 28 30 30 30 30 31 31 32 33 ...
 $ yearc     : num  1918 1918 1918 1918 1918 ...
```

```

$ location: Factor w/ 3 levels "1","2","3": 2 2 1 2 2 2 1 1 1 2 ...
$ bath      : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ kitchen   : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 2 1 1 ...
$ cheating  : Factor w/ 2 levels "0","1": 1 2 2 1 2 2 1 2 1 1 ...
$ district: int   916 813 611 2025 561 541 822 1713 1812 152 ...

```

The output of the function is a new data frame. The results can be seen by using the **R** function `str()`;

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### `data_int2num()`

Occasionally, we need to convert integer variables with a very large range of values into numeric vectors, especially for plots. The function `data_int2num()` performs this conversion.

```
rent99 |> data_int2num() -> da
```

```

*****
      rent  rentsqm    area  yearc location    bath  kitchen cheating
      2723    3053    132    68         3         2         2         2
district
      336
2 integer vectors with more number of values than 50 were transformed to numeric
*****

```

The output of the function is a new data frame.

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### `data_fac2num()`

Occasionally, we need to convert integer variables with a very large range of values into numeric vectors, especially for plots. The function `data_int2num()` performs this conversion.

```
da <- data_fac2num(rent, vars=c("B"))
```

```

*****
*****
the R class of the data is: data.frame
the dimensions of the data before omission are: 1969 x 9

```

```

the dimensions of the data saved after omission are: 1969 x 9
the number of observations omitted: 0
*****
*****

```

```
class(da$B)
```

```
[1] "numeric"
```

The output of the function is a new data frame.

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**data\_rm()**

For analysing datasets, it's easier to keep only the relevant variables. The function `data_rm()` allows you to remove unnecessary variables, making the dataset more manageable for analysis.

```
data_rm(rent99, c(2,9)) -> da
dim(rent99)
```

```
[1] 3082    9
```

```
dim(da)
```

```
[1] 3082    7
```

The output of the function is a new data frame. Note that this could be also done using the function `select()` of the package **dplyr**, or our own `data_select()` function.

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**data\_rmNAvars()**

Occasionally when we read data from files in R some extra variables are accidentally produced with no values but NA's. The function `data_rmNAvars()` removes those variables from the data;

```
data_rmNAvars(da)
```

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**data\_rm1val()**

This function searches for variables with only a single distinct value (often factors left over from a previous `subset()` operation) and removes them from the dataset.

```
da |> data_rm1val()
```

The output of the function is a new data frame.

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**data\_select()**

This function selects specified variables from a dataset.

```
da1<-rent |> data_select( vars=c("R", "Fl", "A"))  
head(da1)
```

|   | R      | Fl | A    |
|---|--------|----|------|
| 1 | 693.3  | 50 | 1972 |
| 2 | 422.0  | 54 | 1972 |
| 3 | 736.6  | 70 | 1972 |
| 4 | 732.2  | 50 | 1972 |
| 5 | 1295.1 | 55 | 1893 |
| 6 | 1195.9 | 59 | 1893 |

The output of the function is a new data frame.

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**data\_exclude\_class()**

This function searches for variables (columns) of a specified `R` class and removes them from the dataset. By default, the class to remove is factor.

```
da |> data_exclude_class() -> da1
head(da1)
```

|   | rent     | area | yearc |
|---|----------|------|-------|
| 1 | 109.9487 | 26   | 1918  |
| 2 | 243.2820 | 28   | 1918  |
| 3 | 261.6410 | 30   | 1918  |
| 4 | 106.4103 | 30   | 1918  |
| 5 | 133.3846 | 30   | 1918  |
| 6 | 339.0256 | 30   | 1918  |

The output of the function is a new data frame.

### **data\_only\_continuous()**

This function keeps only the continuous variables in the dataset, removing all non-continuous variables.

```
da |> data_only_continuous() -> da1
head(da1)
```

|   | rent     | area | yearc |
|---|----------|------|-------|
| 1 | 109.9487 | 26   | 1918  |
| 2 | 243.2820 | 28   | 1918  |
| 3 | 261.6410 | 30   | 1918  |
| 4 | 106.4103 | 30   | 1918  |
| 5 | 133.3846 | 30   | 1918  |
| 6 | 339.0256 | 30   | 1918  |

The output of the function is a new data frame.

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### **data\_factor()**

This function changes the reference level of factors to be the the **lower** or **higher** of the number of observations.

```
da |> gamlss.prepdata::data_factor() -> da1
levels(da1$loc)
```

```
[1] "3" "1" "2"
```

It moved the level 3 into the first place. The output of the function is a new data frame.

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## Graphical functions

Graphical methods are used in pre-analysis to examine individual variables or pair-wise relationships between variables.

Table 2: A summary table of the graphical functions

| Functions                    | Usage                                                                                                                                                    |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>data_plot()</code>     | Plots each variable in the dataset to visualize its distribution or other important characteristics (see also Section                                    |
| <code>data_bucket()</code>   | Generates bucket plots for all numerical variables in the dataset to visualize their skewness and kurtosis                                               |
| <code>data_response()</code> | Plots the response variable alongside its z-score, providing a standardized version of the response for comparison                                       |
| <code>data_zscores()</code>  | Plots the z-scores for all continuous variables in the dataset, allowing for easy visualization of the standardized values (see also Section )           |
| <code>data_xyplot()</code>   | Generates pairwise plots of the response variable against all other variables in the dataset to visualize their relationships                            |
| <code>data_cor()</code>      | Calculates and plots the pairwise correlations for all continuous variables in the dataset to assess linear relationships between them                   |
| <code>data_void()</code>     | Searches for pairwise empty spaces across all continuous variables in the dataset to identify problems with interpretation or prediction                 |
| <code>data_pcor()</code>     | Calculates and plots the pairwise partial-correlations for all continuous variables in the dataset                                                       |
| <code>data_inter()</code>    | Searches for potential pairwise interactions between variables in the dataset to identify relationships or dependencies that may be useful for modelling |



| Functions                       | Usage                                                                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>data_leverage()</code>    | Detects outliers in the continuous explanatory variables (x-variables) as a group to highlight unusual observations                                      |
| <code>data_Ptrans_plot()</code> | Plots the response variable against various power transformations of the continuous x-variables to explore potential relationships and model suitability |

Next we give examples of the graphical functions.

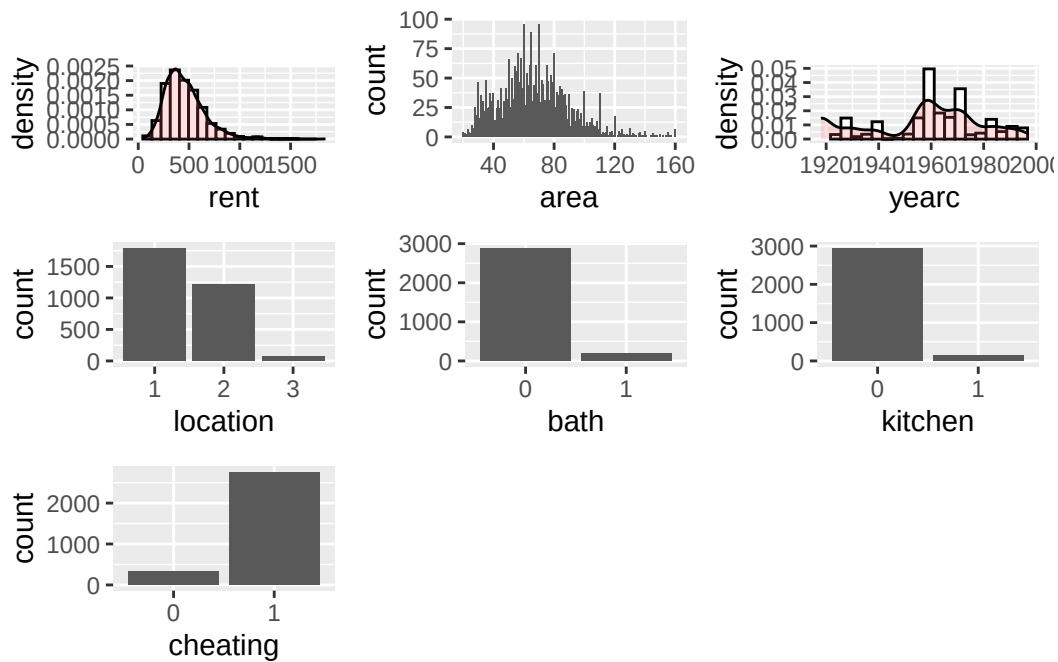
### `data_plot()`

The function `data_plot` plots all the variables of the data individually; It plots the continuous variable as *histograms* with a density plots superimposed, see the plot for `rent` and `yearc`. As an alternative a `dot plots` can be requested see for an example in `?@sec-data_response`. For integers the function plots *needle* plots, see `area` below and for categorical the function plots *bar* plots, see `location`, `bath kitchen` and `cheating` below.

```
da |> data_plot()
```

```
100 % of data are saved,
that is, 3082 observations.
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_bar()`).
Removed 2 rows containing missing values or values outside the scale range
(`geom_bar()`).
```



The function could save the `ggplot2` figures.

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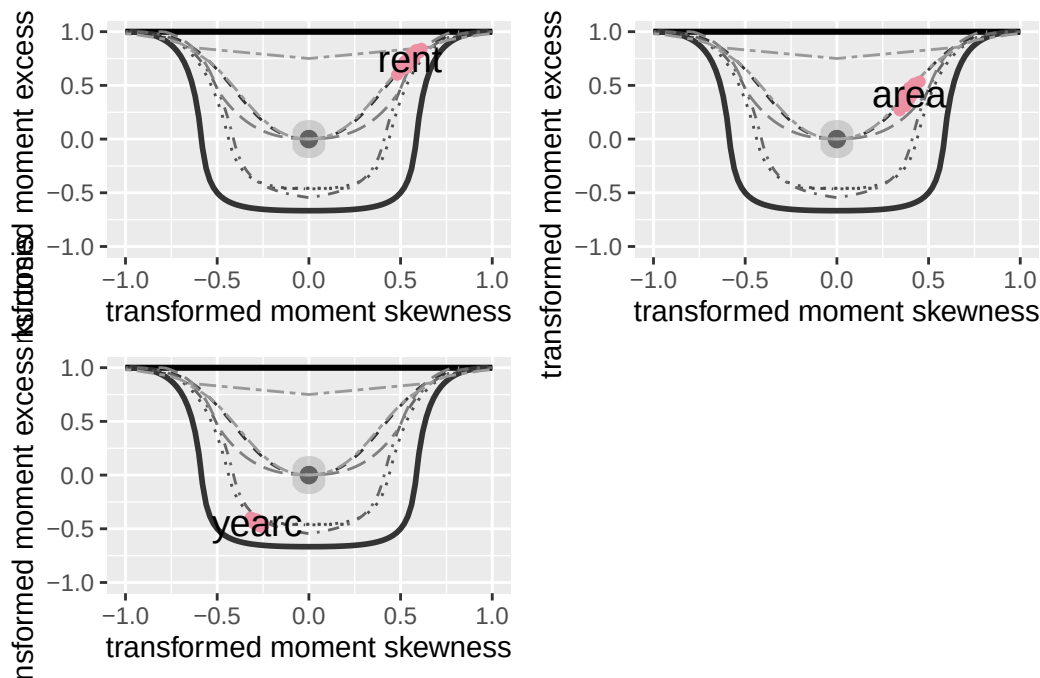
### `data_bucket()`

The function `data_bucket` plots bucket plots (which identify skewness and kurtosis) for all variables in the data set.

```
da |> data_bucket()
```

100 % of data are saved,  
that is, 3082 observations.

| rent | area | yearc | location | bath | kitchen | cheating |
|------|------|-------|----------|------|---------|----------|
| 2723 | 132  | 68    | 3        | 2    | 2       | 2        |



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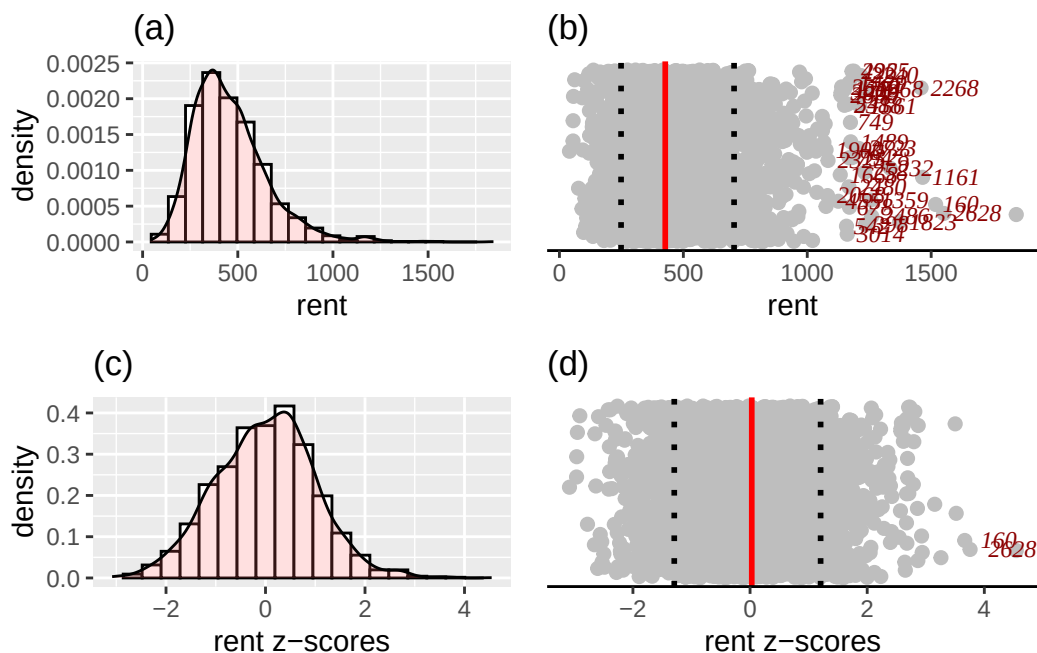
#### `data_response()`

The function `data_response()` plots four different plots related to a continuous response variable. It plots; i) a histogram (and the density function); ii) a dot plot of the response in the original scale iii) a histogram (and the density function) of the response at the z-scores scale and vi) s dot-plot in the z-score scale.

The z-score scale is defined by fitting a distribution to the variable (normal or SHASH) and then taking the residuals, see also next section. The dot plots are good in identify highly skew variables and unusual observations. They display the median and inter quantile range of the data. The y-axis of a dot plot is a randomised uniform variable (therefore the plot could look slightly different each time.)

```
da |> data_response(, response=rent)
```

```
100 % of data are saved,
that is, 3082 observations.
the class of the response is numeric is this correct?
a continuous distribution on (0,inf) could be used
```

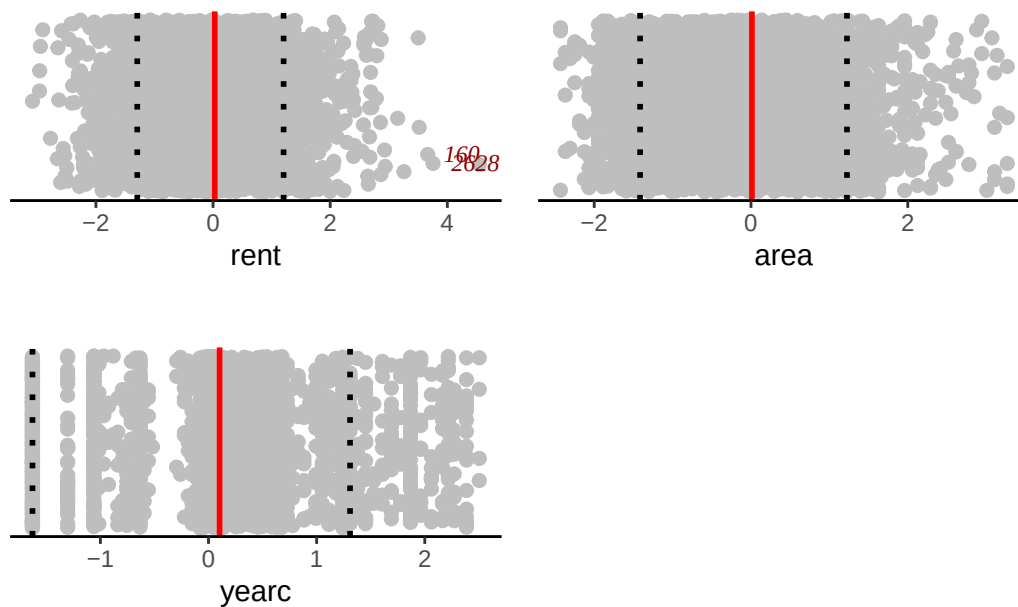


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#### `data_zscores()`

One could fit any four parameter (GAMLSS) distribution, defined on  $-\infty$  to  $\infty$ , to any continuous variable, where skewness and kurtosis is suspected and take the quantile residuals (z-scores) as the transformed values of the x-values. The function `y_zscores()` performs just this. It takes a continuous variable and fits a continuous (four parameter SHASHo) distribution and gives the z-scores. The fitting distribution is specified by the user, but the default is SHASHo. The function `data_zscores()` takes all continuous variables and plots their z-scores. The methodology is use also to identify outliers in `data_outliers()`.

```
da |> data_zscores()
```



In order to see how the z-scores are calculated consider the function `y_zscores()` which is taking an individual variables z-scores;

```
z <- y_zscores(rent99$rent, plot=FALSE)
```

The function is equivalent of fitting a constant model to all the parameters of a given distribution and then taking the quantile residuals (or z=scores) as the variable of interest. The default distribution is the four parameter SHASHo distribution.

```
library(gamlss2)
m1 <- gamlssML(rent99$rent, family=SHASHo) # fitting a 4 parameter distribution
cbind(z,resid(m1))[1:5,]# and taking the residuals
```

```
      z
1 -2.496097 -2.496097
2 -1.340464 -1.340464
3 -1.182014 -1.182014
4 -2.526326 -2.526326
5 -2.295069 -2.295069
```

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`data_xyplot()`

The functions `data_xyplot()` plots the response variable against each of the independent explanatory variables. It plots the continuous against continuous as *scatter* plots and continuous variables against categorical as *box* plot.

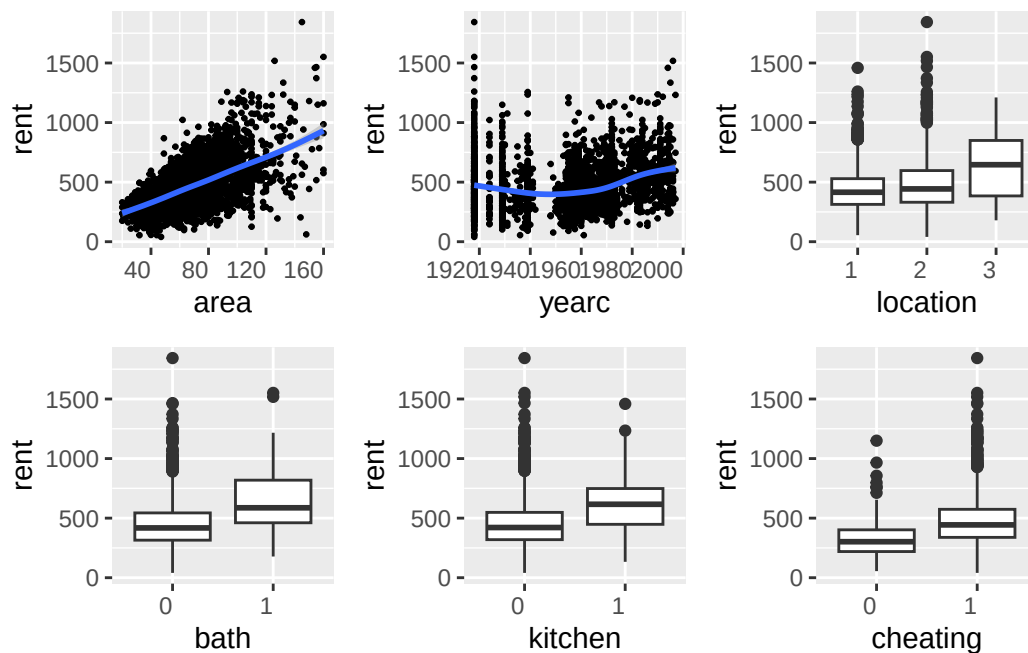
**i** Note

At the moment there is no provision for categorical response variables.

```
da |> data_xyplot(response=rent )
```

100 % of data are saved,  
that is, 3082 observations.

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'  
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



The output of the function saves the `ggplot2` figures.

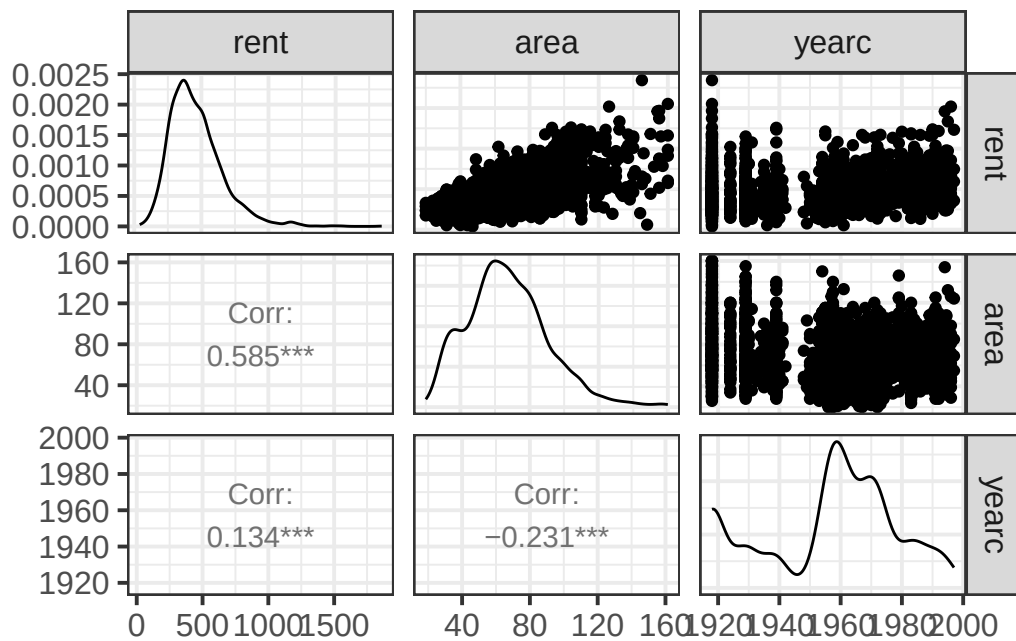
Note that the package `gamlss.prepdata` does not provide pairwise plots of the explanatory variables themselves but the package `GGally` does. Here is an example ;

```
library(GGally)
```

Registered S3 method overwritten by 'GGally':

```
method from  
+.gg ggplot2
```

```
dac <- gamlss.prepdata:::data_only_continuous(da)  
ggpairs(dac, lower=list(continuous = "cor",  
  combo = "box_no_facet", discrete = "count",  
  na = "na"), upper=list(continuous = "points",  
  combo = "box_no_facet", discrete = "count", na = "na")) +  
theme_bw(base_size = 15)
```



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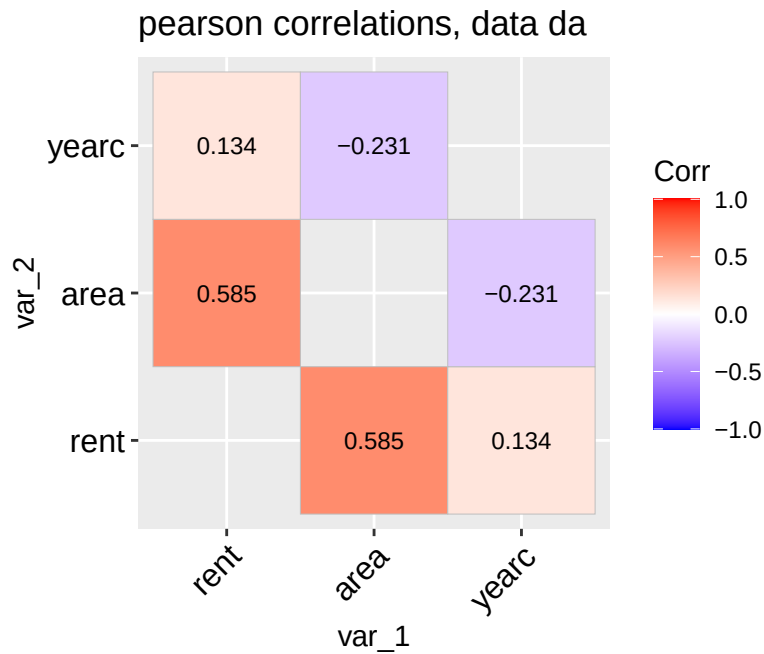
**data\_cor()**

The function `data_corr()` is taking a `data.frame` object and plot the correlation coefficients of all its continuous variables.

```
data_cor(da, lab=TRUE)
```

100 % of data are saved,  
that is, 3082 observations.  
4 factors have been omitted from the data

Warning in data\_cor(da, lab = TRUE):



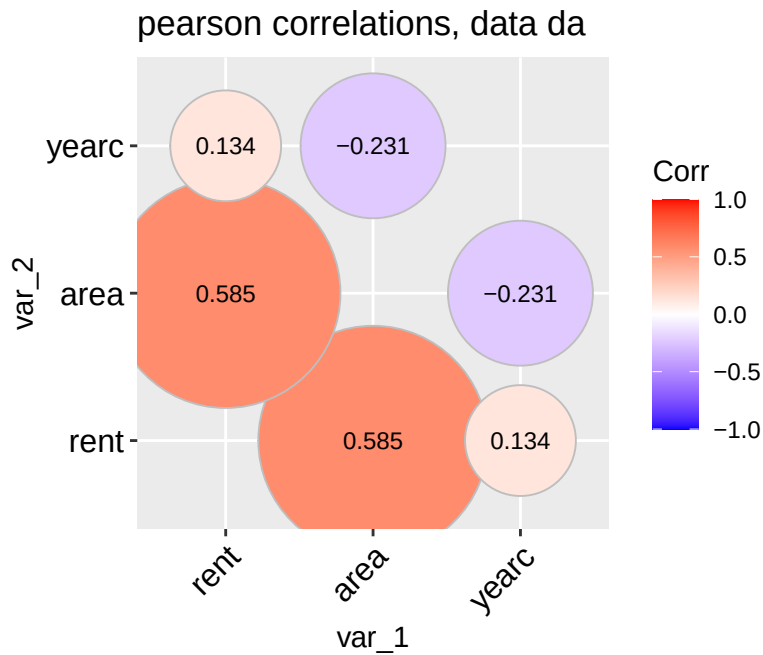
A different type of plot can be produce if we use;

```
data_cor(da, method="circle", circle.size = 40)
```

100 % of data are saved,  
that is, 3082 observations.  
4 factors have been omitted from the data

Warning in data\_cor(da, method = "circle", circle.size = 40):





To get the variables with higher, say, than 0.4 correlation values use;

```
Tabcor <- data_cor(da, plot=FALSE)
```

100 % of data are saved,  
that is, 3082 observations.  
4 factors have been omitted from the data

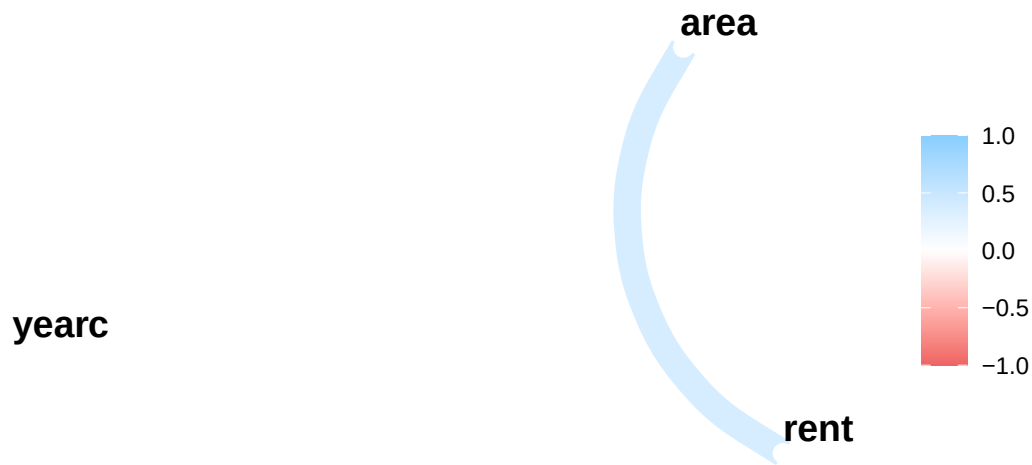
Warning in data\_cor(da, plot = FALSE):

```
high_val(Tabcor, val=0.4)
```

```
      name1 name2 corrs
[1,] "rent" "area" "0.585"
```

We can plot the path of those variables using the package `corr`;

```
library(corr)
network_plot(Tabcor)
```



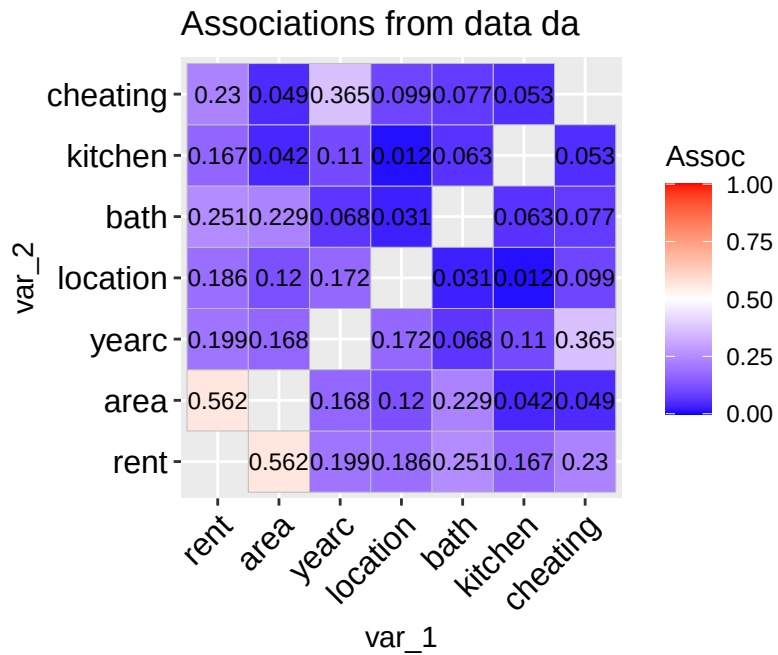
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#### `data_association()`

The function `data_association()` is taking a `data.frame` object and plot the pair-wise association of all its variables. The pair-wise association for two continuous variables is given by default by the absolute value of the Spearman's correlation coefficient. For two categorical variables by the (adjusted for bias) Cramers'  $v$ -coefficient. For a continuous against a categorical variables by the  $\sqrt{R^2}$  obtained by regressing the continuous variable against the categorical variable.

```
data_association(da, lab=TRUE)
```

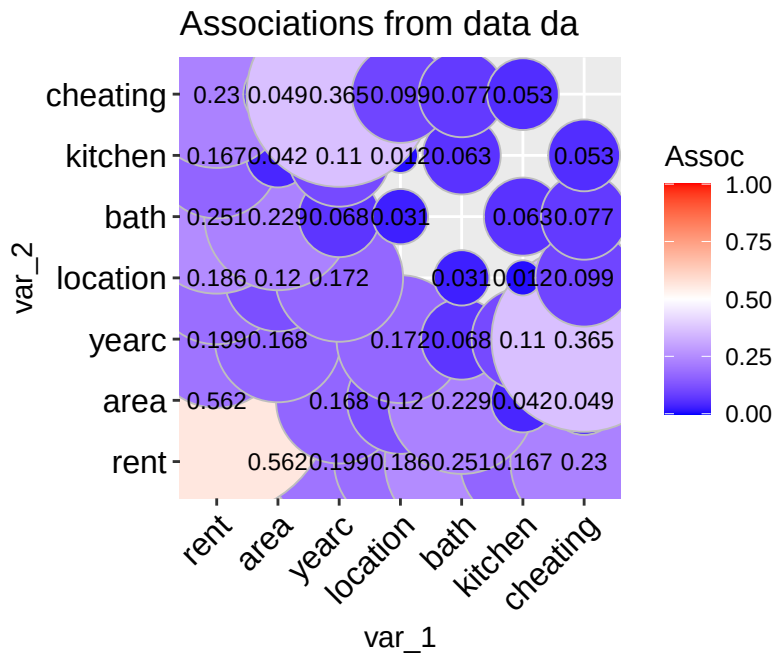
100 % of data are saved,  
that is, 3082 observations.



A different type of plot can be produce if we use;

```
data_association(da, method="circle", circle.size = 40)
```

100 % of data are saved,  
that is, 3082 observations.



To get the variables with higher, say, than 0.4 association values use;

```
Tabcor <- data_cor(da, plot=FALSE)
```

100 % of data are saved,  
that is, 3082 observations.  
4 factors have been omitted from the data

Warning in data\_cor(da, plot = FALSE):

```
high_val(Tabcor, val=0.4)
```

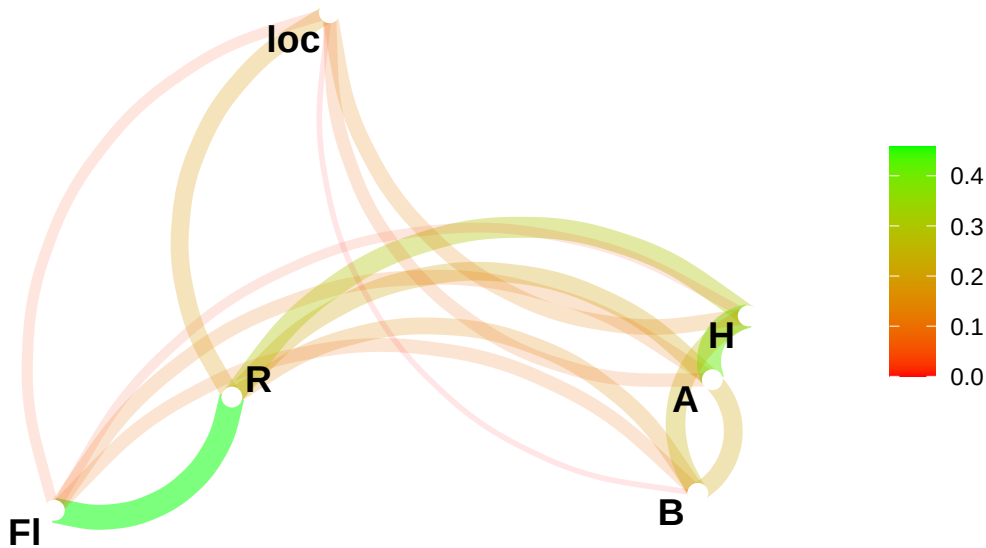
```
      name1 name2 corrs
[1,] "rent" "area" "0.585"
```

A different plot can be achieved using the function `network_plot()` of package `corr`;

```
pp=gamlss.prepdata::data_association(rent[, -c(4,5,8)], plot=F)
```

100 % of data are saved,  
that is, 1969 observations.

```
library(corr)
network_plot(pp, min_cor = 0, colors = c("red", "green"), legend = "range")
```



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**data\_void()**

#### ⚠ Warning

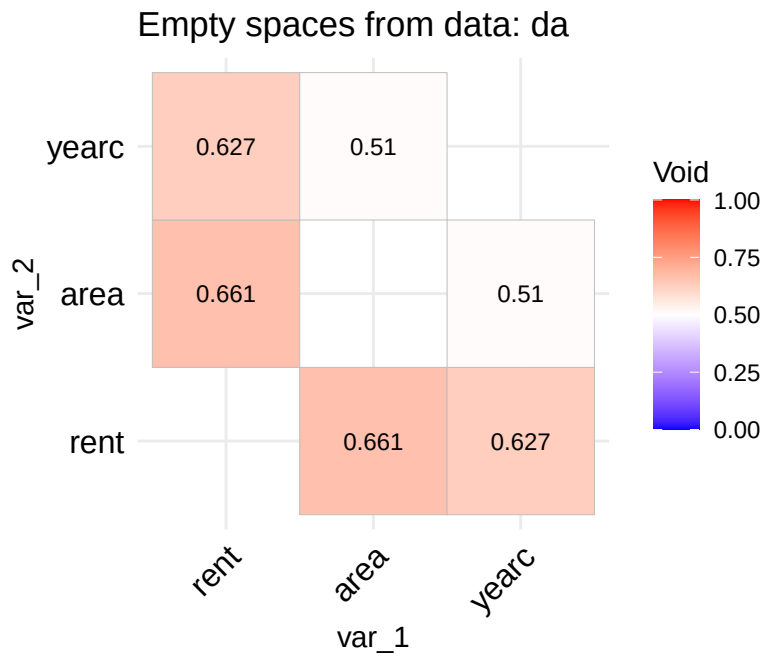
This function is new. Its theoretical foundation are not proven yet. The function needs testing and therefore it should be used with caution.

The idea behind the functions `void()` and its equivalent `data.frame` version `data_void()` is to be able to identify whether the data in the direction of two continuous variables say  $x_i$  and  $x_j$  have a lot of empty spaces. The reason is that empty spaces effect prediction since interpolation at empty spaces is dangerous. The function `data_void()` is taking a `data.frame` object and plot the percentage of empty spaces for all pair-wise continuous variables. The function used the `foreach()` function of the package **foreach** to allow parallel processing.

```
registerDoParallel(cores = 9)
data_void(da)
```

```
100 % of data are saved,
that is, 3082 observations.
4 factors have been omitted from the data
```

Warning in data\_void(da):

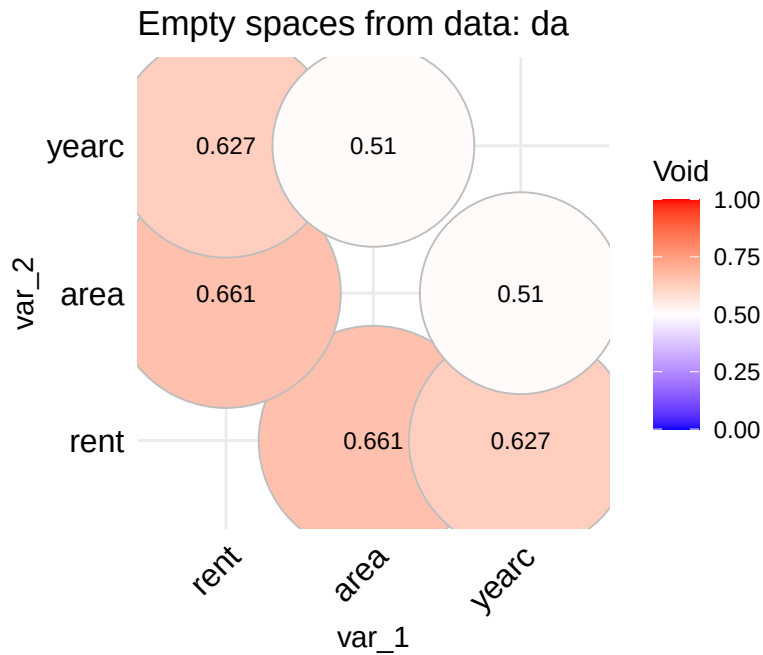


A different type of plot can be produce if we use;

```
data_void(da, method="circle", circle.size = 40)
```

100 % of data are saved,  
that is, 3082 observations.  
4 factors have been omitted from the data

Warning in data\_void(da, method = "circle", circle.size = 40):



```
stopImplicitCluster()
```

To get the variables with higher than 0.4 values use;

```
Tabvoid <- data_void(da, plot=FALSE)
```

100 % of data are saved,  
that is, 3082 observations.  
4 factors have been omitted from the data

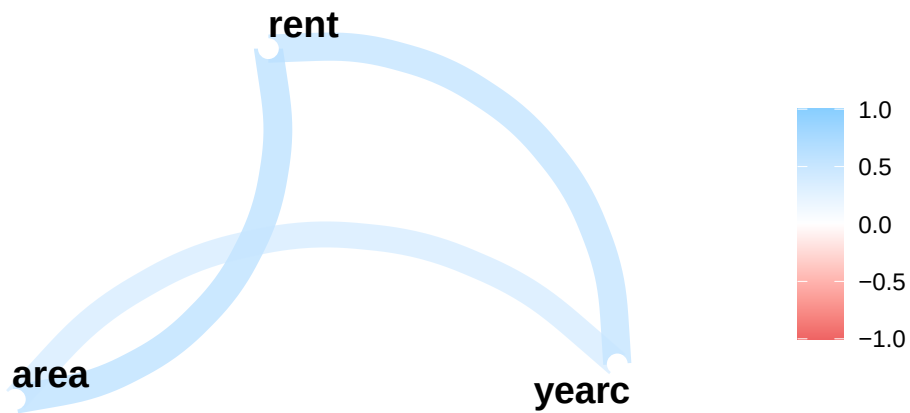
Warning in data\_void(da, plot = FALSE):

```
high_val(Tabvoid, val=0.4)
```

|      | name1  | name2   | corrs   |
|------|--------|---------|---------|
| [1,] | "rent" | "area"  | "0.661" |
| [2,] | "rent" | "yearc" | "0.627" |
| [3,] | "area" | "yearc" | "0.51"  |

To plot their paths use;

```
library(corr)
network_plot(Tabvoid)
```



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`data_pcor()`

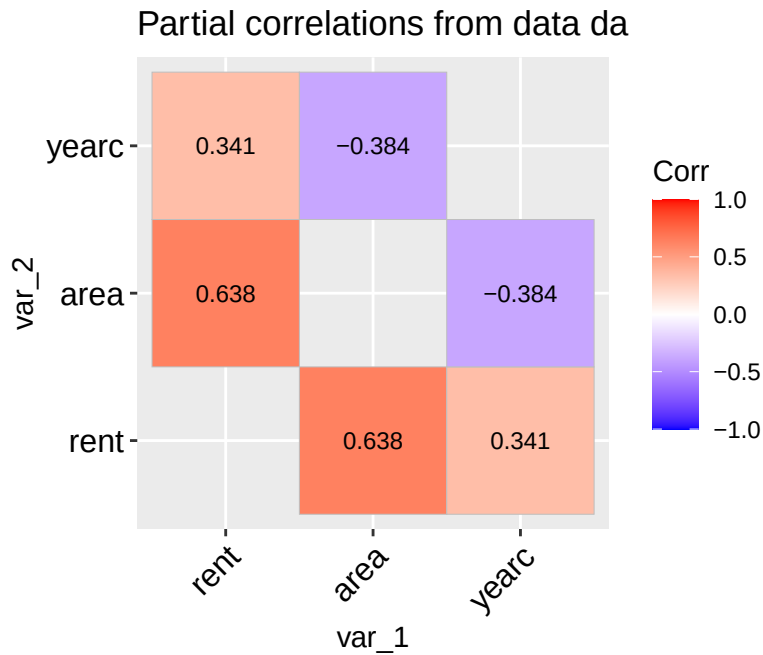
The function `data_copr()` is taking a `data.frame` object and plot the partial correlation coefficients of all its continuous variables.

```
data_pcor(da, lab=TRUE)
```

```
100 % of data are saved,  
that is, 3082 observations.  
4 factors have been omitted from the data
```

```
Warning in data_pcor(da, lab = TRUE):
```





To get the variables with higher than 0.4 partial correlation values use;

```
Tabpcor <- data_pcor(da, plot=FALSE)
```

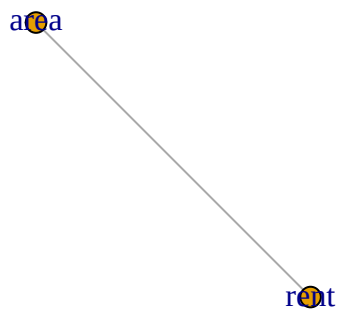
100 % of data are saved,  
that is, 3082 observations.  
4 factors have been omitted from the data

Warning in data\_pcor(da, plot = FALSE):

```
high_val(Tabpcor, val=0.4)
```

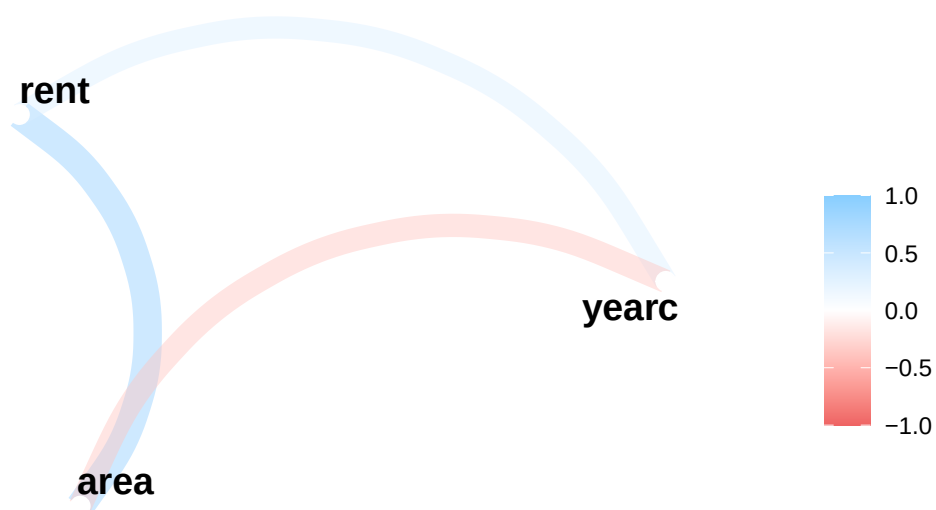
```
      name1 name2 corrs
[1,] "rent" "area" "0.638"
```

```
high_val(Tabpcor, val=0.4, plot=T)
```



For plotting you can use;

```
library(corrnet)
network_plot(Tabpcor)
```



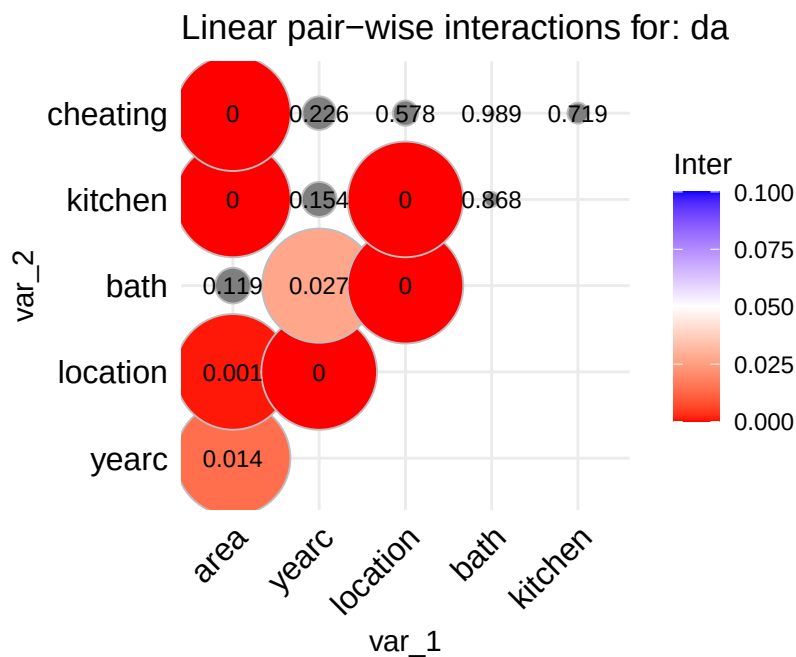
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### `data_inter()`

The function `data_inter()` takes a `data.frame`, fits all pair-wise interactions of the explanatory variables against the response (using a normal model) and produce a graph displaying their significance levels. The idea behind this is to identify possible first order interactions at an early stage of the analysis.

```
da |> gamlss.prepdata::data_inter(response= rent)
```

100 % of data are saved,  
that is, 3082 observations.



```
tinter <- gamlss.prepdata:::data_inter(da, response= rent, plot=FALSE)
```

100 % of data are saved,  
that is, 3082 observations.

```
tinter
```

|          | area | yearc | location | bath  | kitchen | cheating |
|----------|------|-------|----------|-------|---------|----------|
| area     | NA   | 0.014 | 0.001    | 0.119 | 0.000   | 0.000    |
| yearc    | NA   | NA    | 0.000    | 0.027 | 0.154   | 0.226    |
| location | NA   | NA    | NA       | 0.000 | 0.000   | 0.578    |
| bath     | NA   | NA    | NA       | NA    | 0.868   | 0.989    |
| kitchen  | NA   | NA    | NA       | NA    | NA      | 0.719    |
| cheating | NA   | NA    | NA       | NA    | NA      | NA       |

To get the variables with lower than \$0.05 \$ significant interactions use;

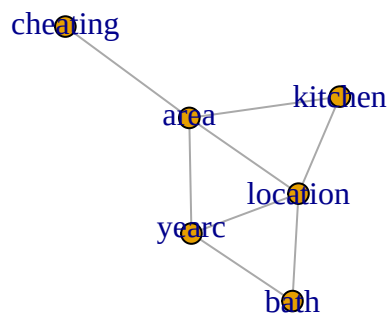
```
Tabinter <- gamlss.prepdata:::data_inter(da, response= rent, plot=FALSE)
```

100 % of data are saved,  
that is, 3082 observations.

```
low_val(Tabinter)
```

|      | name1      | name2      | value   |
|------|------------|------------|---------|
| [1,] | "area"     | "yearc"    | "0.014" |
| [2,] | "area"     | "location" | "0.001" |
| [3,] | "yearc"    | "location" | "0"     |
| [4,] | "yearc"    | "bath"     | "0.027" |
| [5,] | "location" | "bath"     | "0"     |
| [6,] | "area"     | "kitchen"  | "0"     |
| [7,] | "location" | "kitchen"  | "0"     |
| [8,] | "area"     | "cheating" | "0"     |

```
low_val(Tabinter, plot=T)
```



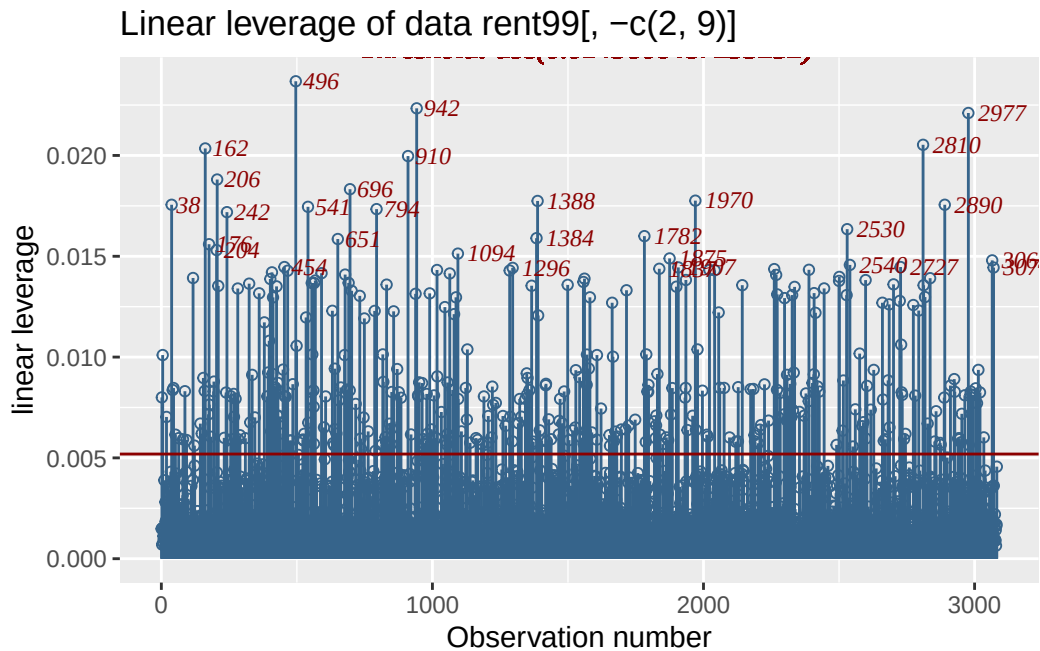
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### **data\_leverage()**

The function `data_leverage()` uses the linear model methodology to identify possible unusual observations within the explanatory variables as a group (not individually). It fit a linear (normal) model with response, the response of the data, and all explanatory variables in the data, as x's. It then calculates the leverage points and plots them. A leverage is a number between zero and 1. Large leverage correspond to extremes in the x's.

```
rent99[, -c(2,9)] |> data_leverage( response=rent)
```

100 % of data are saved,  
that is, 3082 observations.



**i** Note

The horizontal line is the plot is at point  $2 \times (r/n)$ , which is the threshold suggested in the literature; values beyond this point could be identified as extremes. It looks that the point  $2 \times (r/n)$  is too low (at least for our data). Instead the plot identify only observation which are in the one per cent upper quantile. That is a quantile value of the leverage at value 0.99.

Section show how the information of the function `data_leverage()` can be combined with the information given by `data_outliers()` in order to confirm high outliers in the data.

**!** Important

What happen if multiple response are in the data?

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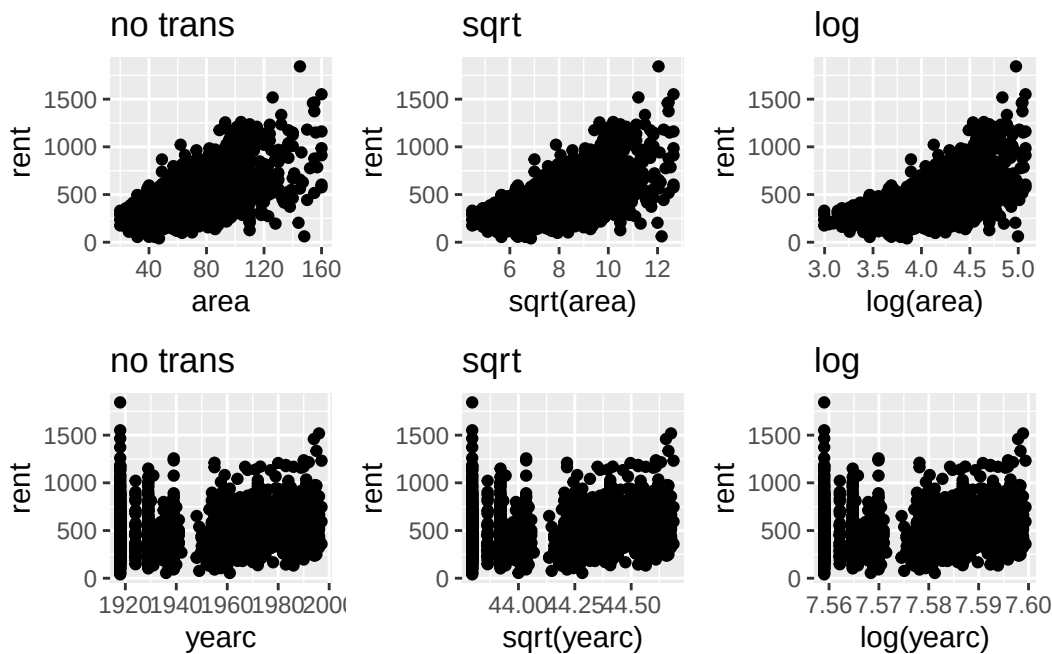
**data\_Ptrans\_plot()**

The essence of this plot is to visually decide whether certain values of  $\lambda$  in a power transformation are appropriate or not. Section describes the power transformation,  $T = X^\lambda$  for  $\lambda > 0$ , in more details. For each continuous explanatory variable  $X$ , the function show the response against  $X$ , ( $\lambda = 1$ ), the response against the square root of  $X$ , ( $\lambda = 1/2$ ) and the response against the log of  $X$  ( $\lambda \rightarrow 0$ ). The user then can decide whether any of those transformation

are appropriate. Note that the square root and log transformations are appropriate for right skew variables.

```
gamlss.prepdata::data_Ptrans_plot(da, rent)
```

100 % of data are saved,  
that is, 3082 observations.



It look that no transformation is needed for the continuous explanatory variables **area** of **yearc**.

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## Feature functions

The features functions are functions to help with the continuous explanatory x-variables; We divide them here in five sections; Functions for

- identify outliers;
- identify high correlation between variables;
- scaling continuous variables;
- transforming continuous variables and

- function appropriate for `dates` and `time` (helpful for time series data analysis).

The available functions are;

Table 3: The outliers, scaling, transformation and date and time functions

| Functions                             | Usage                                                                                                                                                                     |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>y_outliers()</code>             | identify possible outliers for one continuous variable                                                                                                                    |
| <code>y_outliers_both()</code>        | identify possible outliers using both z-scores and the quantile rule methodology                                                                                          |
| <code>y_outliers_loop()</code>        | identify outliers (using z-scores) by fitting several times the chosen <code>family</code> eliminating each time observations identified as outliers in the previous fits |
| <code>y_outliers_by()</code>          | identify possible outliers (using z-scores) at partitions of the data by a factor                                                                                         |
| <code>y_outliers_z()</code>           | identify possible outliers (using z-scores) for different combination of <code>loop</code> and <code>by</code>                                                            |
| <code>data_outliers()</code>          | identify possible outliers for all continuous x-variable (see also <code>data_zscores</code> in Section )                                                                 |
| <code>data_leverage()</code>          | this is a repeat of the Section function                                                                                                                                  |
| <code>data_index_cor()</code>         | this function gives an index of continuous variables with high correlation with others continuous variables                                                               |
| <code>data_index_association()</code> | this function gives an index of variables with high association with others continuous or factor variables                                                                |
| <code>data_scale()</code>             | scales all continuous x-variables to zero mean and one s.d. or to the range [0,1]                                                                                         |
| <code>xy_Ptrans()</code>              | looking for appropriate power transformation for response against one of the x-variable                                                                                   |
| <code>data_Ptrans()</code>            | looking for appropriate power transformation for response against all x-variable                                                                                          |
| <code>data_Ptrans_plot()</code>       | plotting all x's against the response using identity, square root and log transformations                                                                                 |
| <code>time_dt2dh()</code>             | Separates a date and time variable into two variables                                                                                                                     |
| <code>time_hour2num()</code>          |                                                                                                                                                                           |

## Outliers

Outliers in the response variable, within a distributional regression framework like GAMLSS, are better handled by selecting an appropriate distribution for the response. For example, an outlier under the assumption of a normal distribution for the response may no longer be considered an outlier if a distribution from the t-family is assumed. Considerations of contaminated responses data are not considered here. The function `data_response()`, discussed in Section and uses the z-scores methodology, provides some guidelines for appropriate action to identify outliers in the response. This section focuses on outliers in the explanatory variables.

Outliers in the continuous explanatory variables can potentially affect curve fitting, whether using linear or smooth non-parametric terms. In such cases, removing outliers from the x-variables can make the model more robust. Outliers are observations that deviate significantly from the rest of the data, but the concept also depends on the dimension being considered. An single observation value might be an outlier on an explanatory variable. A pair of observations could be an outlier when examined within a pair-wise relationships. Extreme observations identification on continuous variables, is based on how far the observation lies from the majority of the data. Pairwise plots of continuous variables could identify outliers in the two-dimensions. Leverage points are useful to identify outliers in  $r$  dimensions where  $r$  is the size of the explanatory variables.

### Note

At this preliminary stage of the analysis (no model has been fitted yet), it is difficult to identify outliers among **factors**. This is because outliers in factors typically do not appear unusual unless are examined in combination with rest of factors or variables. Identifying **factor** outliers is a task better suited for the modelling stage of the analysis.

The function `y_outliers()` in Section , is applied to continuous variables only, and it uses two different methodologies (option `type`);

- the z-scores, `type="zscorsces"` (the default) and
- the quantile rule, `type="quantile"`.

Both methodologies are trying to remove extreme skewness in the variables before looking for outliers by using the (internal) function `y_Ptrans()`. The function is applied to any variable with positive values. The assumption, is that positive variables are more likely to be right skewed and therefor a power transformation will correct that. The function `y_Ptrans()` attempts to find the power  $\lambda$ , of a transformation  $T = X^\lambda$ , that minimizes the Jarque-Bera test statistic—a measure of deviation from normality. The Jarque-Bera test statistic is always non-negative and a value way from zero indicates strong evidence against normality. The goal of this methodology is to reduce extreme skewness in the data before attempt to identify the outliers. If the data are not so skewed the power transformation will genraly do no harm.



#### **i** Note

At the moment we allow the power parameter of the transformation  $\lambda$  to vary at the range 0 to 1.5. This range covers only right skewed behaviour. More work has to be done to establish a suitable range.

Now given that the positive variables are transformed (using the power transformation) and that all the rest of variables, containing negative values, remain untransformed, the z-scores methodology fits a parametric distribution to the variable and computes the residuals of the fitted model (z-scores). The default distribution is the four parameter **SHASHo** distribution. The z-scores methodology identifies outliers if a specific residual (z-score) has an absolute value bigger than a predetermined value  $K$ . The quantile rule methodology works similarly. It identifies an observation if it is further than  $K \times MAD$  away from the median. The value of  $K$  is set by the option `value` of the function `y_outliers()`. If the `value` is missing it is calculated using the formula  $\Phi^{-1}(1/(10 \times N))$ , where  $N$  is the number of observations. The two methodologies are appropriate for identifying outliers in one dimension. To identify outliers in  $p$  dimensions the function `data_leverage()`, see Section , should be used.

#### **i** Note

Several of the graphical functions described in Section are good for identifying outliers visually. In fact we recommend that the function `data_outliers()` and its related function `data_outliers_both()`, `data_outliers_by()` and `data_outliers_z()` should be used in combination with functions like `data_plot()` or `data_zscores()` or `y_dots()` of the package `gamlss.gplots`.

### **y\_outliers()**

The function `y_outliers()` identifies outliers in one  $x$  variable using the two methodologies described above. The

- **zscores**: where a distribution is fitted to the data (default **SHASHo**), the residuals (z-scores) are taken and observations with large residuals are identified as outliers.
- **quantile**: This is more in line with the classical methodology based on sample quantiles. An observation is identified if it is far from the median. How far is usually taken as  $K$  times the Mean Absolute Deviation (MAD).

Here is an example using the zscores;

```
y_outliers(da$rent)
```

the  $x$  was transformed using the power 0.2412793

```
named integer(0)
```

and here using the `quantile`;

```
y_outliers(da$rent, type="quantile")
```

```
the x was transformed using the power 0.2412793
```

```
integer(0)
```

Note that the function `y_outliers()` is used by `data_outliers()` to identify outliers in all continuous variables in the data. There are several variation of the basic function `y_outliers()` described next.

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```
y_outliers_both()
```

The function `y_outliers_both()` uses both `zscores` and `quantile` methodology and reports the observations appearing in the intersection of both sets, `method=intersection` (the default), or the union of both, `method=union`.

```
y_outliers_both(da$rent, method="union")
```

```
the x was transformed using the power 0.2412793
```

```
the x was transformed using the power 0.2412793
```

```
integer(0)
```

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```
y_outliers_by()
```

The function `y_outliers_by()` splits the data according to given factor, option `by` before it performs the detection of outliers. It is useful if the given factor split the data in a meaningful way. Continuous variable can be split in advance to create a factors using the **R** function `cut()`.

```
y_outliers_by(da$rent, by=da$location)
```

```
$`1`  
[1] 1747
```

```
$`2`  
[1] 243
```

```
$`3`  
numeric(0)
```

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**y\_outliers\_loop()**

The function `y_outliers_loop()` tries to identify more data than the `y_outliers()` function by using the following extra loop; It starts by fitting the distribution to the data which help to identify from the z-scores certain observations as outliers. Those outliers are then removed and the process starts again. It stops where there are no more observations removed. This loop allows more observations to be identified and it is designed to help when a bunch of data points masks other points to be identified as outliers.

```
gamlss.prepdata:::y_outliers_loop(da$rent)
```

the x was transformed using the power 0.2412793

```
integer(0)
```

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**y\_outliers\_z()**

The function `y_outliers_z()` tries to put the functions `y_outliers_loop()` and `y_outliers_by()` together. It works only for the z-scores methodology but its options allow different combinations of the algorithm. For example the options;

- **by**: allows to fit for levels of a factor;
- **value**: allows different values in the detection of z-scores outliers;
- **family**; allows different families of distribution to be fitted (instead of the **SHASHo**)

- `transform` allows to switch off the transformation option
- `loop`; allows looping the algorithm, `TRUE` or `FALSE`.

```
gamlss.prepdata::y_outliers_z(da$rent, by=da$loc)
```

```
the x was transformed using the power 0.2412793
```

```
$`1`  
integer(0)
```

```
$`2`  
[1] 38
```

```
$`3`  
integer(0)
```

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### `data_outliers()`

The function `data_outliers()` uses the *z-scores* technique described above in order to detect outliers in the continuous variables in the data. It fits a **SHASHo** distribution to the continuous variable in the data and uses the z-scores (quantile residuals) to identify (one dimension) outliers for those continuous variables.

```
data_outliers(da)
```

```
the x was transformed using the power 0.2412793  
the x was transformed using the power 0.3872234  
the x was transformed using the power 1.499942
```

```
$rent  
named integer(0)
```

```
$area  
named integer(0)
```

```
$yearc  
named integer(0)
```

As it happen no individual variable outliers were highlighted in the `rent` data using the z-scores methodology. In general we recommend the function `data_outliers()` to be used in combination of the graphical functions `data_plot()` and `data_zscores()`

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### `data_leverage()` (repeat)

The function `data_leverage()`, first describe in Section , can be used to detect extremes in the continuous variables in the data by fitting a linear model with all the continuous variables in the data and then using the higher leverage points to identify (multi-dimension) outliers for the observations for all continuous variables. By default, it identifies one percent of the observations with high leverage in the data. Here we use the function as a way to identify the observation (not plotting).

```
data_leverage(da, response=rent, plot=FALSE)
```

100 % of data are saved,  
that is, 3082 observations.

```
[1] 38 162 176 204 206 242 454 496 541 651 696 794 910 942 1094  
[16] 1296 1384 1388 1782 1837 1875 1907 1970 2530 2540 2727 2810 2890 2977 3065  
[31] 3070
```

#### Note

The function `data_leverage()` always identifies the top one percent of observations with the highest leverage, which may or may not be outliers. These high-leverage points can have a strong influence on model estimates. To better assess their impact, the list of high-leverage observations returned by `data_leverage()` should be compared with the list of outliers identified by the `data_outliers()` function. Observations that appear in both lists are particularly noteworthy, as they may be influential outliers that warrant further investigation.

Next we are trying the process of identifying if data with high leverage are also coincide with outliers identified using the function `data_outliers()` function. The function `intersect()` will flash out the intersection of the two set of observations. First use `data_outliers()`

```
ul <- unlist(data_outliers(da))
```

```
the x was transformed using the power 0.2412793
the x was transformed using the power 0.3872234
the x was transformed using the power 1.499942
```

```
then data_leverage();
```

```
l1<-data_leverage(da, response=rent, plot=FALSE)
```

```
100 % of data are saved,
that is, 3082 observations.
```

and finally we inspect the intersection of the two sets using `intersect()`;

```
intersect(l1,u1)
```

```
integer(0)
```

Here we have zero outliers but in general we would expect to find common observations for further checking.

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## High correlations

The idea here is to identify variables with high correlation or association with the possibility to eliminate them from the data. The first function is appropriate for continuous variables while the second for both continuous and factors. Note that the method used to identify the variables is identical to the method used in the package `caret` see page 5 of Kuhn and Max (2008).

```
data_index_cor()
```

The function `data_index_cor()`, It takes a `data.frame` as input calculates correlation coefficients of the continuous variables in the data and creates an index for variables which could be possible to eliminate from the further analysis.

```
gamlss.prepdata::data_index_cor(rent99[,-1])
```

```
4 factors have been omitted from the data
```

```
Warning in data_cor(data, type = type, plot = FALSE, percentage = percentage, :  
numeric(0)
```

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#### `data_index_association()`

The function `data_index_association()`, It takes a `data.frame` as input calculates the association coefficients of all the variables in the data and creates an index for variables which could be possible to eliminate from the further analysis.

```
gamlss.prepdata:::data_index_association(rent99[, -1])
```

```
numeric(0)
```

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## Scaling

Scaling is another form of transformation for the continuous explanatory variables in the data which brings the explanatory variables in the same scale. It is a form of **standardization**. Standardization means bringing all continuous explanatory variables to similar range of values. For some machine learning techniques, i.e., principal component regression or neural networks standardization is mandatory in other like LASSO is recommended. There are two types of standardisation

- i) *scaling to normality* and
- ii) *scaling to a range from zero to one*.

Scaling to normality it means that the variable should have a mean zero and a standard deviation one. Note that, this is equivalent of fitting a Normal distribution to the variables and then taking the z-scores (residuals) of the fit as the transformed variable. The problem with this type of scaling is that skewness and kurtosis in the standardised data persist. One could go further and fit a four parameter distribution instead of a normal where the two extra parameters could account for skewness and kurtosis. The function `data_scale()` described in Section it gives this option with the argument `family` in which can be used to specify a different distribution.

`data_scale()`

The function `data_scale()` perform scaling to all continuous variables an a data set. Note that factors in the data set are left untouched because when fitted within a model they will be transform to dummy variable which take values 0 or 1 (a kind of standardization). The response is left also untouched because it is assumed that an appropriate distribution will be fitted. The response variable has to be declared using the argument `response`. First we use `data_scale()` to scale to normality.

```
head(data_scale(da, response=rent))
```

|  | area | yearc | location | bath | kitchen | cheating |
|--|------|-------|----------|------|---------|----------|
|  | 132  | 68    | 3        | 2    | 2       | 2        |

|   | rent     | area       | yearc      | location | bath | kitchen | cheating |
|---|----------|------------|------------|----------|------|---------|----------|
| 1 | 109.9487 | 0.7009962  | 0.7035881  | 2        | 0    | 0       | 0        |
| 2 | 243.2820 | -0.4796117 | 0.7484206  | 2        | 0    | 0       | 1        |
| 3 | 261.6410 | -0.2266243 | 0.6587556  | 1        | 0    | 0       | 1        |
| 4 | 106.4103 | -0.3531180 | -1.2242096 | 2        | 0    | 0       | 0        |
| 5 | 133.3846 | 0.1950214  | -1.7173671 | 2        | 0    | 0       | 1        |
| 6 | 339.0256 | -0.5639408 | 1.6450707  | 2        | 0    | 0       | 1        |

As we mention before scaling to normality is equivalent of fitting a Normal variable first. The problem with scaling to normality is that if the variables are highly skew or kurtotic scaling them to normality does correct for skewness or kurtosis. Using a parametric distribution with four parameters some of which are skewness and kurtosis parameters it may correct that. Next we standardised using the SHASHo distribution;

```
head(data_scale(da, response=rent, family="SHASHo"))
```

|  | area | yearc | location | bath | kitchen | cheating |
|--|------|-------|----------|------|---------|----------|
|  | 132  | 68    | 3        | 2    | 2       | 2        |

|   | rent     | area       | yearc      | location | bath | kitchen | cheating |
|---|----------|------------|------------|----------|------|---------|----------|
| 1 | 109.9487 | 0.7390434  | 0.6323846  | 2        | 0    | 0       | 0        |
| 2 | 243.2820 | -0.3940193 | 0.6795499  | 2        | 0    | 0       | 1        |
| 3 | 261.6410 | -0.1175129 | 0.5864434  | 1        | 0    | 0       | 1        |
| 4 | 106.4103 | -0.2526941 | -1.0598183 | 2        | 0    | 0       | 0        |
| 5 | 133.3846 | 0.2949483  | -1.6281459 | 2        | 0    | 0       | 1        |
| 6 | 339.0256 | -0.4916986 | 2.0610405  | 2        | 0    | 0       | 1        |



The second form of standardisation is to transform all continuous x-variables to a range zero to one. Here is how this is done;

```
head(data_scale(da, response=rent, scale.to="0to1"))
```

|  | area | year | location | bath | kitchen | cheating |
|--|------|------|----------|------|---------|----------|
|  | 132  | 68   | 3        | 2    | 2       | 2        |

|   | rent     | area       | year | location | bath | kitchen | cheating |
|---|----------|------------|------|----------|------|---------|----------|
| 1 | 109.9487 | 0.04285714 | 0    | 2        | 0    | 0       | 0        |
| 2 | 243.2820 | 0.05714286 | 0    | 2        | 0    | 0       | 1        |
| 3 | 261.6410 | 0.07142857 | 0    | 1        | 0    | 0       | 1        |
| 4 | 106.4103 | 0.07142857 | 0    | 2        | 0    | 0       | 0        |
| 5 | 133.3846 | 0.07142857 | 0    | 2        | 0    | 0       | 1        |
| 6 | 339.0256 | 0.07142857 | 0    | 2        | 0    | 0       | 1        |

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## Transformations

In the context of outliers, the focus is on how far a single observation deviates from the rest of the data. However, when considering **transformations**, the goal shifts toward a better capturing of the relationship between a continuous predictor and the response variable. Specifically, we aim to model **peaks and troughs** in the relationship between a single x-variable and the response better. To achieve this, it is sometimes sufficient to stretch the x-axis so that the predictor variable is more evenly distributed within its range. A common approach for this is to use a power transformation of the form  $T = X^\lambda$ . This transformation can reduce the curvature in the response and make the curve fitting easier. Importantly, the power transformation smoothly transitions into a logarithmic transformation as  $\lambda \rightarrow 0$ , since:  $\frac{X^\lambda - 1}{\lambda} \rightarrow \log(X)$  as  $\lambda \rightarrow 0$ . Thus, in the limit as  $\lambda \rightarrow 0$  the power transformation is the log transformation  $T = \log(X)$ .

### **i** Note

The power transformation is a subclass of the shifted power transformation  $T = (\alpha + X)^\lambda$  which is not considered here

The aim of the power transformation is to make the model fitting process more robust and reliable and maximizing the information extracted from the data. There are three different functions in `gamlss.prepdata` dealing with power transformations, `xy_Ptrans()`, `data_Ptrans()` and `data_Ptrans_plot()`

### **i** Note

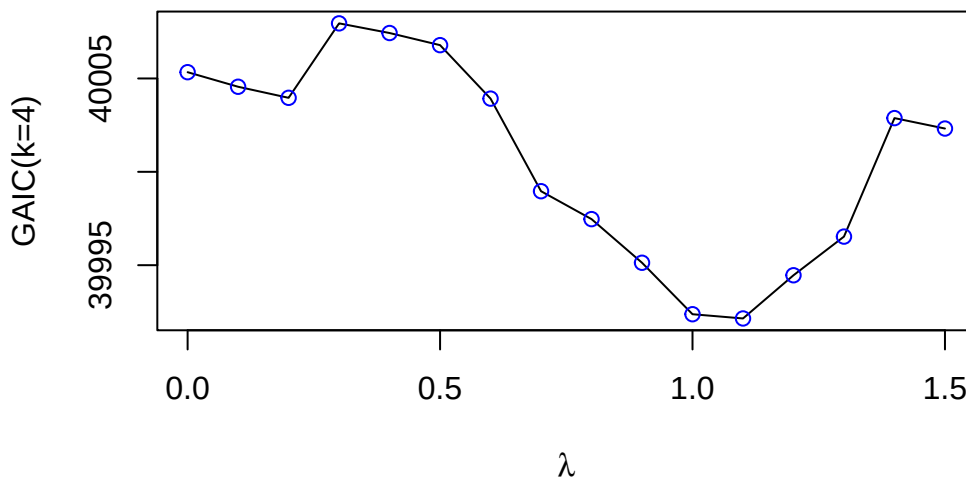
It's important to distinguish between the concepts of *transformations* (the subject of this section) and of *feature extraction*, a term often used in machine learning. We refer to transformations as functions applied to a individual explanatory variables while we use feature extraction when multiple explanatory variables are involved. In both cases, the goal is to enhance the model capabilities.

```
xy_Ptrans()
```

```
pp<-gamlss.prepdata:::xy_Ptrans(da$area, da$rent)  
pp
```

```
[1] 1.06212
```

```
gamlss.prepdata:::xy_Ptrans(da$area, da$rent, prof=TRUE, k=4)
```



```
data_Ptrans()
```

```
gamlss.prepdata:::data_Ptrans(da, response=rent)
```

```
100 % of data are saved,  
that is, 3082 observations.
```

```
$area  
[1] 1.06212
```

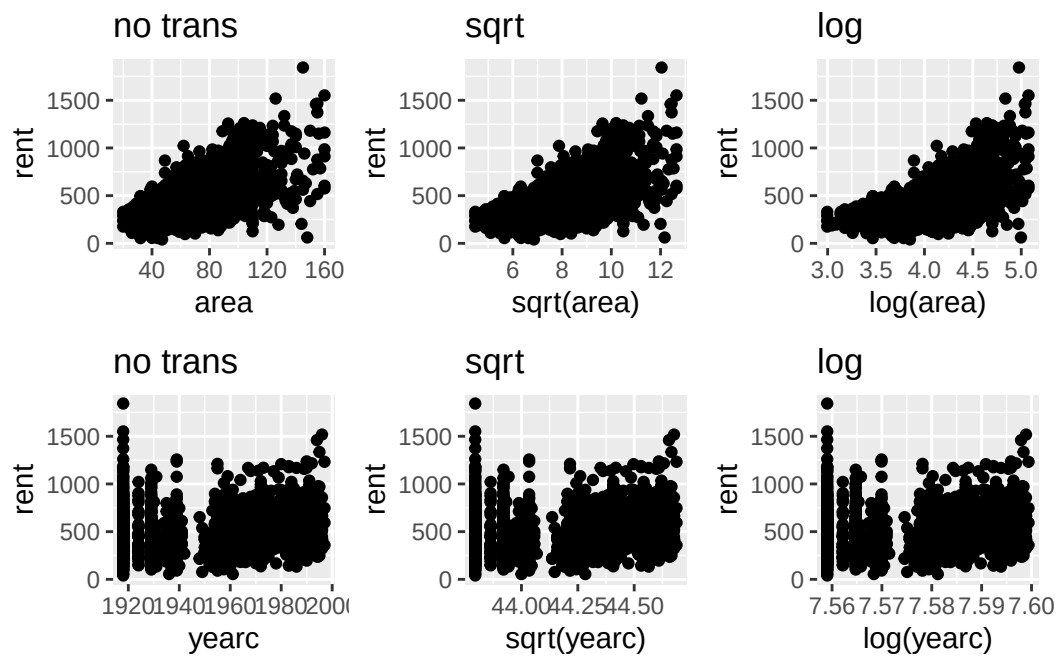
```
$yearc  
[1] 1.499942
```

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**data\_Ptrans\_plot() (repeat)**

```
gamlss.prepdata:::data_Ptrans_plot(da, rent)
```

100 % of data are saved,  
that is, 3082 observations.



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## Dates and Time

Time series data sets consist of observations collected at consistent time intervals—e.g., daily, weekly, or monthly. These data require special treatment because observations that are closer

together in time tend to be more similar, violating the usual assumption of independence between observations. A similar issue arises in spatial data sets, where observations that are geographically closer often exhibit spatial correlation, again violating independence.

Many data sets include spatial or temporal features —such as longitude and latitude or timestamps (e.g., 12/10/2012)— those variables are can be used for for modeling directly, but also for interpreting or visualization of the results. If temporal or spatial information exists in the data, it is essential to ensure these features are correctly read and interpreted in R. For instance, dates in R can be handled using the `as.Date()` function. To understand its functionality and options associated with the function please consult the help documentation via `help("as.Date")`. Below, we provide two simple functions of how to work with date-time variables.

#### `time_dt2dh()`

Suppose the variable `dt` contains both a date and a time. We may want to extract and separate this into two components: one containing the date, and another capturing the hour of the day.

```
dt <- c("01/01/2011 00:00", "01/01/2011 01:00", "01/01/2011 02:00", "01/01/2011 03:00", "01/01/2011 04:00", "01/01/2011 05:00")
```

The function to do this separation can look like;

```
time_dt2dh <- function(datetime, format=NULL)
{
  X <- t(as.data.frame(strsplit(datetime, ' ')))
  rownames(X) <- NULL
  colnames(X) <- c("date", "time")
  hour <- as.numeric(sub(":", ".", X[,2]))
  date <- as.Date(X[,1], format=format)
  data.frame(date, hour)
}
```

```
P <- time_dt2dh(dt, "%d/%m/%Y")
P
```

|   | date       | hour |
|---|------------|------|
| 1 | 2011-01-01 | 0    |
| 2 | 2011-01-01 | 1    |
| 3 | 2011-01-01 | 2    |
| 4 | 2011-01-01 | 3    |
| 5 | 2011-01-01 | 4    |
| 6 | 2011-01-01 | 5    |

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`time_hour2num()`

The second example contains time given in its common format, `c("12:35", "10:50")` and we would like `t` change it to numeric which we can use in the model.

```
t <- c("12:35", "10:50")
```

The function to change this to numeric;

```
time_hour2num <- function(time, pattern=":")
{
  t <- gsub(pattern, ".", time)
  as.numeric(t)
}
time_hour2num(t)
```

```
[1] 12.35 10.50
```

For user who deal with a lot of time series try the function `POSXct()` and also the function `select()` from the `dplur` package.

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## Data Partition

### Introduction to data partition

Partitioning the data is essential for comparing different models and checking for overfitting. Overfitting occurs when a model fails to generalize well to new data, typically because it is too closely fitted to the data. In contrast, underfitting refers to a poor model that fails to capture the essential patterns in both fitted and the new data, resulting to a model that is far the population intended to represent. It common to refer to data used for fitting the model as **training** data and the new data as **test** data.

Figure 1 shows different ways the data can be partitioned. A dataset can be split once, in **holdout** samples, or split multiple times using cross validation, **CV**, or **bootstrap**. For large datasets, it is common to split the data into a *training* set and two holdout samples, the *test* set, and, if needed the *validation* set. The training set is used for fitting the models, the test set is used to evaluate the prediction power and also to compare models, and the validation

set is used for tuning the **hyperparameters**. [Note that for additive smoothing regression models, the hyper parameters are the **smoothing** parameters.] Observations in the training set are often referred to as the **in-bag** data, while observations in the test or validation set are termed the **out-of-bag** data.

Partitioning the data into training, test, and validation sets should be done randomly. This brings us to the two possible problems which could occur when splitting the data; The **zero variance variable** and the **extreme values** problems. The zero variance variable (or nearly zero variance problem) occurs when a variable in the data (could be a factor) has only few distinct values and the frequency of one of them is very low. For example, if we have the factor **gender** with mostly **female** values and only few **male** and we randomly partitioning the data, there is a good chance that all males could end up in only one of the partitions. That is, one of the partitions would have factor gender properly with two levels, while the other partition with only one level. This would make the fitting and interpretation of the model difficult. One should check if this is the case. The extreme values problem is similar in nature but will effect overfitting. Extreme values could occur in both the explanatory and the response variables. For the first case we have deal with **outliers** in Section we concentrate here on extreme values in the response. Extreme values in the response (if they genuine) are of great importance for risk analysis. There is a danger that such events when splitting the data can go to one partition and not to the other. Any evaluation of risks in this case could be potentially wrong. One possible solution could be **repeated** partition, i.e bootstrap.

Bootstrap, middle of 1, is a repeated partition method. In the classical, non-parametric bootstrap, the index of the data i.e.  $1, 2, \dots, n$  is sampled with replacement  $B$  times and the model is refitted on the re-sample data. In the Bayesian bootstrap the data are weighted using weighted samples from a multinomial distribution.  $B$  re-weighted models are refitted. While both classical and Bayesian bootstraps are computationally expensive, they provide additional insights into the model and in particular the variability of all of its parameters. A sample of length  $B$  is obtained for all the parameters of interest.

In  $K$ -fold cross-validation, (right part of Figure 1), the data are split into  $K$  sub-samples. The model is then fitted  $K$  times, each time using  $K - 1$  sub-samples for training and the remaining one sub-sample as test data set. By the end of the process, all  $n$  observations will have been used as **test** data exactly once. This is the main advantage of  $K$ -fold cross-validation is that provides reliable test data for model evaluation, (with the cost of fitting  $K$  models). Leave-One-Out Cross-Validation (LOO) is a special case of  $K$ -fold cross-validation, where the model is trained on  $n - 1$  observations and tested on the remaining single observation, iterating this process  $n$  times. Consequently,  $n$  different models are refitted.

**i** Note

Repeated partitions (i.e. bootstrapping and k-fold cross validation) could mitigate some of the problems associated from a single partitions.

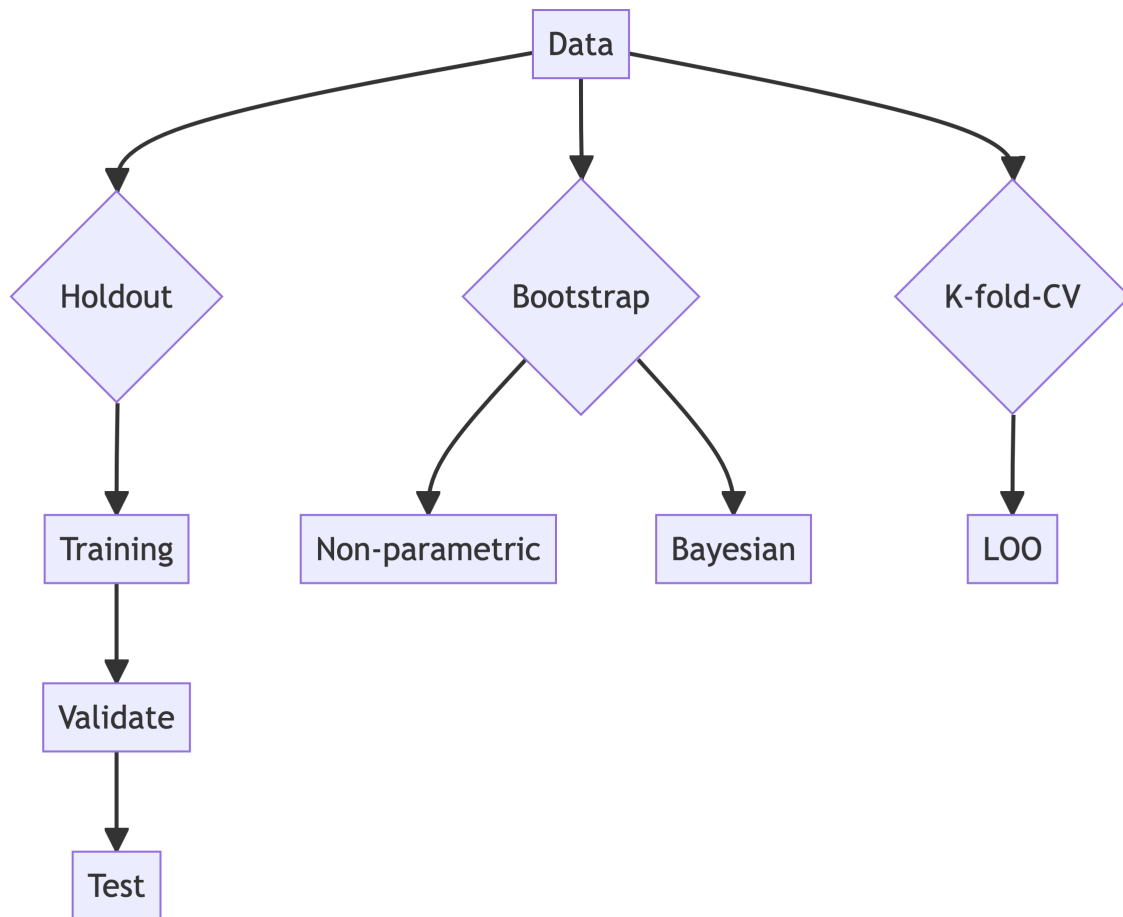


Figure 1: Different ways of splitting data to get in order to get more information”

Data partitioning help in selection of models, model diagnostics and detection of overfitting. Here are some rules of thumb applied to distributional regression models;

- if the residuals from the training data and the test data sets behave well, one is confident, that the distributional assumption is adequate.
- if the fitted model performs reasonably well (using a measure of goodness-of-fit) on both the training and the test data sets, one can be confident that overfitting has been avoided.

All methods in Figure 1, need performance metrics. Those metrics can help can: i) facilitate model comparison, ii) detect overfitting, and iii) select hyper-parameters.

Various goodness-of-fit measures can be used to check and compare different distributional regression models. One of the most popular methods which do not required partition of the data is the Generalized Akaike Information Criterion (GAIC), see Akaike (1983). To calculate the GAIC, only the in-bag data are required. However, calculating the GAIC requires an accurate measure of the degrees of freedom used to fit the model. For mathematical (stochastic) models, the degrees of freedom are straight forward to define as they correspond to the number of estimated parameters. In contrast, for over-parametrised algorithmic models like neural networks, estimating the degrees of freedom is challenging. Note that the degrees of freedom serve as a measure of the model's complexity. We expect more complex models to be closer to the training data, which may reduce their ability to generalize well to new data (overfitting).

#### **i** Note

Note that the most common of goodness-of-fit measures used in machine learning models requiring data partition are **not** appropriate for distributional regression model like GAMLSS. This because those measures concentrate in the behaviour at the centre of the distribution of the response which may or not be the focus of the study.

The most common measures of goodness-of-fit used in machine learning models with partitioning are;

- *RMSE*: root mean squared error;
- *MAE*: mean absolute error;
- $R^2$ : and
- *MAPE*: mean absolute percentage error.

All those measures of goodness-of-fit are OK if someone is interested how the middle part of the distribution of the response is behaving but are NOT good if the interest is in the tail or if the distribution is very skew. For distributional regression model the following are more appropriate measures of goodness-of-fits.

- the **log-likelihood**: i.e.  $\ell = \log L$ , or equivalent the **deviance**  $d = -2\ell$



- *CRPS*; the continuous rank probability scores.

Note that in general goodness-of-fit measure could be can applied to both the `in bag` and `out of bag` data sets;

Computationally partition of the data for any distributional regression model, can be achieved in different ways;

1. by *splitting* the original data frame in different subsets;
2. by using a *factor* which classifies the partitions of data sets;
3. by *indexing* the original data, i.e. `data[index,]`,
4. by using *prior weights* in the fitting.

So while we would like to think data partitioning as cutting of the original data into different subset (case 1 above), there is no inherent need for physically partition into the subgroups when fitting the models since the same results can be achieved with different methods. For example using prior weights in a  $K$ -fold cross-validation, we need  $K$  dummy vectors  $i_k$ . The dummy vectors take the value of 1 for the  $k$ -th set of observations and 0 for the rest.

The function `data_part()` creates a factor for holdout or k-fold CV partitions. The `data_part_list()` creates a list with different `data.frames` for holdout or k-fold CV partitions. The function `data_boot_index()` and `data_Kfold_index()` creates appropriate vector for indexing, while `data_boot_weights()` and `data_Kfold_weights()` generates the appropriate vectors for prior weights.

Table 4: The data partition functions

| Functions                         | Usage                                                                                                  |
|-----------------------------------|--------------------------------------------------------------------------------------------------------|
| <code>data_part()</code>          | Creates a single or multiple (CV) partitions by introducing a factor with different levels             |
| <code>data_part_list()</code>     | Creates a single or multiple (CV) partitions with output a list of <code>data.frames</code>            |
| <code>data_boot_index()</code>    | Creates two lists. The in-bag, IB, indices for fitting and the out-of-bag, OOB, indices for prediction |
| <code>data_boot_weights()</code>  | Create a $n \times B$ matrix with columns weights for bootstrap fits                                   |
| <code>data_Kfold_index()</code>   | Creates a $n \times K$ matrix of variables to be used for cross validation data indexing               |
| <code>data_Kfold_weights()</code> | Creates a $n \times K$ matrix of dummy variables to be used for cross validation weighted fits         |

| Functions               | Usage                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------|
| <code>data_cut()</code> | This is not a partition function but randomly select a specified proportion of the data |

Here are the data partition functions.

### `data_part()`

The function `data_part()` it does not partitioning the data as such but adds a factor called `partition` to the data indicating the partition. By default the data are partitioned into two sets the training ,`train`, and the test data, `test`.

```
dap <- gamlss.prepdata:::data_part(da)
```

data partition into two sets

```
head(dap)
```

|   | rent     | area | yearc | location | bath | kitchen | cheating | partition |
|---|----------|------|-------|----------|------|---------|----------|-----------|
| 1 | 109.9487 | 26   | 1918  | 2        | 0    | 0       | 0        | train     |
| 2 | 243.2820 | 28   | 1918  | 2        | 0    | 0       | 1        | test      |
| 3 | 261.6410 | 30   | 1918  | 1        | 0    | 0       | 1        | train     |
| 4 | 106.4103 | 30   | 1918  | 2        | 0    | 0       | 0        | test      |
| 5 | 133.3846 | 30   | 1918  | 2        | 0    | 0       | 1        | test      |
| 6 | 339.0256 | 30   | 1918  | 2        | 0    | 0       | 1        | train     |

If the user would like to split the data in two separate `data.frame`'s she/he can use;

```
daTrain <- subset(dap, partition=="train")
daTest <- subset(dap, partition=="test")
dim(daTrain)
```

```
[1] 1846    8
```

```
dim(daTest)
```

```
[1] 1236    8
```

Note, that use of the option `partition` has the following behaviour;

- i) `partition=2L` (the default) the factor has two levels `train`, and `test`.
- ii) `partition=3L` the factor has three levels `train`, `val` (for validation) and `test`.
- iii) `partition > 4L` say `K` then the levels are `1, 2...K`. The factor then can be used to identify `K`-fold cross validation, see also the function `data_Kfolds_CV()`.

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### `data_part_list()`

The function `data_part_list()` creates a list of `dataframes` which can be used either for single partition or for cross validation. The argument `partition` allows the splitting of the data to up to 20 subsets. The default is a list of 2 with elements `training` and `test` (single partition).

Here is a single partition in `train` and `test` data sets.

```
allda <- data_part_list(rent)
length(allda)
```

```
[1] 2
```

```
dim(allda[["train"]]) # training data
```

```
[1] 1200    9
```

```
dim(allda[["test"]]) # test data
```

```
[1] 769    9
```

Here is a multiple partition for cross validation.

```
allda <- data_part_list(rent, partition=10)
```

10-fold data partition

```
length(allda)
```

```
[1] 10
```

```
names(allda)
```

```
[1] "1" "2" "3" "4" "5" "6" "7" "8" "9" "10"
```

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```
data_boot_index()
```

The function `data_boot_index()` create two lists. The first called IB (for in-bag) and the second OOB, (out-of-bag). Each element of the list IB can be use to select a bootstrap sample from the original `data.frame` while each element of OOB can be use to select the out of sample data points. Note that each bootstrap sample approximately contains  $2/3$  of the original data so we expect on average to  $1/3$  to be in OOB sample.

```
DD <- data_boot_index(rent, B=10)
```

The class, length and names of the created lists;

```
class(DD)
```

```
[1] "list"
```

```
length(DD)
```

```
[1] 2
```

```
names(DD)
```

```
[1] "IB" "OOB"
```

the first observations of the

```
head(DD$IB[[1]]) # in bag
```

```
[1] 1 2 4 6 8 8
```

```
head(DD$OOB[[1]]) # out-of-bag
```

```
[1] 3 5 7 15 21 22
```

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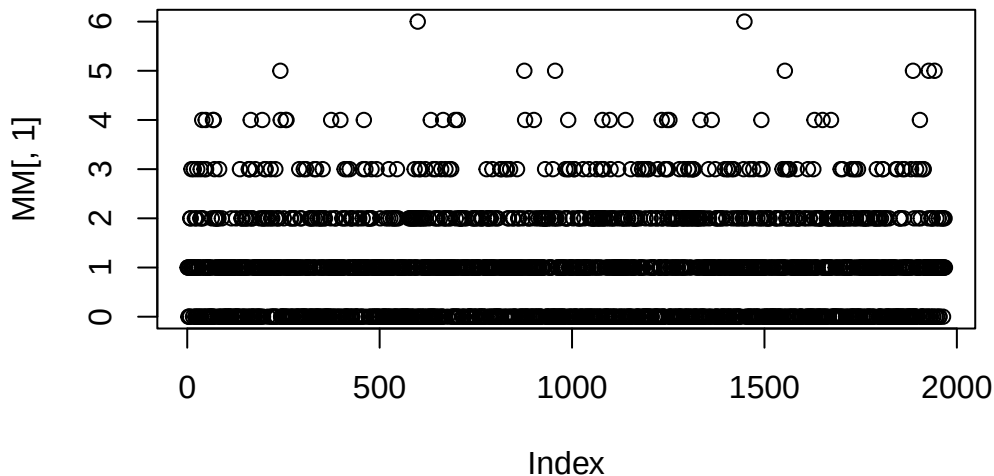
**data\_boot\_weights()**

The function `data_boot_weights()` create a  $n \times B$ . matrix with columns possible weights for bootstrap fits. Note that each bootstrap sample approximately contains .666 of the original obsbation so we in general we expect 0.333 of the data to have zero weights.

```
MM <- data_boot_weights(rent, B=10)
```

The M is a matrix which columns can be used as weights in bootstrap fits; Here is a plot of the first column

```
plot(MM[,1])
```



and here is the first 6 rows os the matrix;

```
head(MM)
```

|      | [,1] | [,2] | [,3] | [,4] | [,5] | [,6] | [,7] | [,8] | [,9] | [,10] |
|------|------|------|------|------|------|------|------|------|------|-------|
| [1,] | 1    | 2    | 2    | 1    | 0    | 1    | 3    | 1    | 3    | 1     |
| [2,] | 1    | 1    | 0    | 2    | 2    | 1    | 2    | 1    | 1    | 2     |
| [3,] | 0    | 0    | 2    | 1    | 2    | 1    | 2    | 1    | 1    | 0     |
| [4,] | 1    | 0    | 1    | 0    | 3    | 0    | 1    | 1    | 0    | 2     |
| [5,] | 0    | 2    | 3    | 1    | 1    | 1    | 2    | 1    | 2    | 1     |
| [6,] | 1    | 0    | 0    | 1    | 0    | 1    | 4    | 0    | 1    | 0     |

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```
data_Kfold_index()
```

The function `data_Kfold()` creates a matrix which columns can be use for cross validation either as indices to select data for fitting or as prior weights.

```
PP <- data_Kfold_index(rent, K=10)
head(PP)
```

|      | [,1] | [,2] | [,3] | [,4] | [,5] | [,6] | [,7] | [,8] | [,9] | [,10] |
|------|------|------|------|------|------|------|------|------|------|-------|
| [1,] | 1    | 1    | 1    | 1    | 0    | 1    | 1    | 1    | 1    | 1     |
| [2,] | 2    | 2    | 0    | 2    | 2    | 2    | 2    | 2    | 2    | 2     |
| [3,] | 3    | 3    | 3    | 3    | 3    | 3    | 3    | 3    | 0    | 3     |
| [4,] | 4    | 4    | 4    | 4    | 4    | 0    | 4    | 4    | 4    | 4     |
| [5,] | 5    | 5    | 5    | 5    | 0    | 5    | 5    | 5    | 5    | 5     |
| [6,] | 6    | 6    | 6    | 6    | 6    | 6    | 6    | 0    | 6    | 6     |

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```
data_Kfold_weights()
```

The function `data_Kfold()` creates a matrix which columns can be use for cross validation either as indices to select data for fitting or as prior weights.

```
PP <- data_Kfold_weights(rent, K=10)
head(PP)
```

|      | [,1] | [,2] | [,3] | [,4] | [,5] | [,6] | [,7] | [,8] | [,9] | [,10] |
|------|------|------|------|------|------|------|------|------|------|-------|
| [1,] | 1    | 1    | 1    | 1    | 0    | 1    | 1    | 1    | 1    | 1     |
| [2,] | 1    | 1    | 0    | 1    | 1    | 1    | 1    | 1    | 1    | 1     |
| [3,] | 1    | 1    | 1    | 1    | 1    | 1    | 1    | 1    | 0    | 1     |
| [4,] | 1    | 1    | 1    | 1    | 1    | 0    | 1    | 1    | 1    | 1     |
| [5,] | 1    | 1    | 1    | 1    | 0    | 1    | 1    | 1    | 1    | 1     |
| [6,] | 1    | 1    | 1    | 1    | 1    | 1    | 1    | 0    | 1    | 1     |

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### `data_cut()`

The function `data_cut()` is not included in the partition Section for its data partitioning properties. It is designed to select a random subset of the data specifically for plotting purposes. This is especially useful when working with large datasets, where plotting routines—such as those from the `ggplot2` package—can become very slow.

The `data_cut()` function can either:

- Automatically reduce the data size based on the total number of observations, or
- Subset the data according to a user-specified percentage.

Here is an example of the function assuming that that the user requires 50 of the data;

```
da1 <- data_cut(rent99, percentage=.5)
```

50 % of data are saved,  
that is, 1541 observations.

```
dim(rent99)
```

```
[1] 3082    9
```

```
dim(da1)
```

```
[1] 1541    9
```

The function `data_cut()` is used extensively in many plotting routines within the `gamlss.ggplots` package. When the percentage option is not specified, a default rule is applied to determine the proportion of the data used for plotting. This approach balances performance and visualization quality when working with large datasets.

Let  $n$  denote the total number of observations. Then:

- if  $n \leq 50,000$ : All data are used for plotting.
- if  $50,000 < n \leq 100,000$ : 50% of the data are used.
- If  $100,000 < n \leq 1,000,000$ : 20% of the data are used.
- If  $n > 1,000,000$  : 10% of the data are used.

This default behaviour ensures that plotting remains efficient even with very large datasets.

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## Distribution Families

### **i** Note

All functions listed below belong to the `gamlss.ggplots` package, not the `gamlss.prepdata` package. However, since they can be useful during the pre-modeling stage, they are presented here. The later version of the `gamlss.ggplots` can be found in <https://github.com/gamlss-dev/gamlss.ggplots>

In the following function, no fitted model is required—only values for the parameters need to be specified.

### `family_pdf()`

The function `family_pdf()` plots individual probability density functions (PDFs) from distributions in the `gamlss.family` package. Although it typically requires the `family` argument, it defaults to the Normal distribution (NO) if none is provided.

```
family_pdf(from=-5,to=5, mu=0, sigma=c(.5,1,2))
```



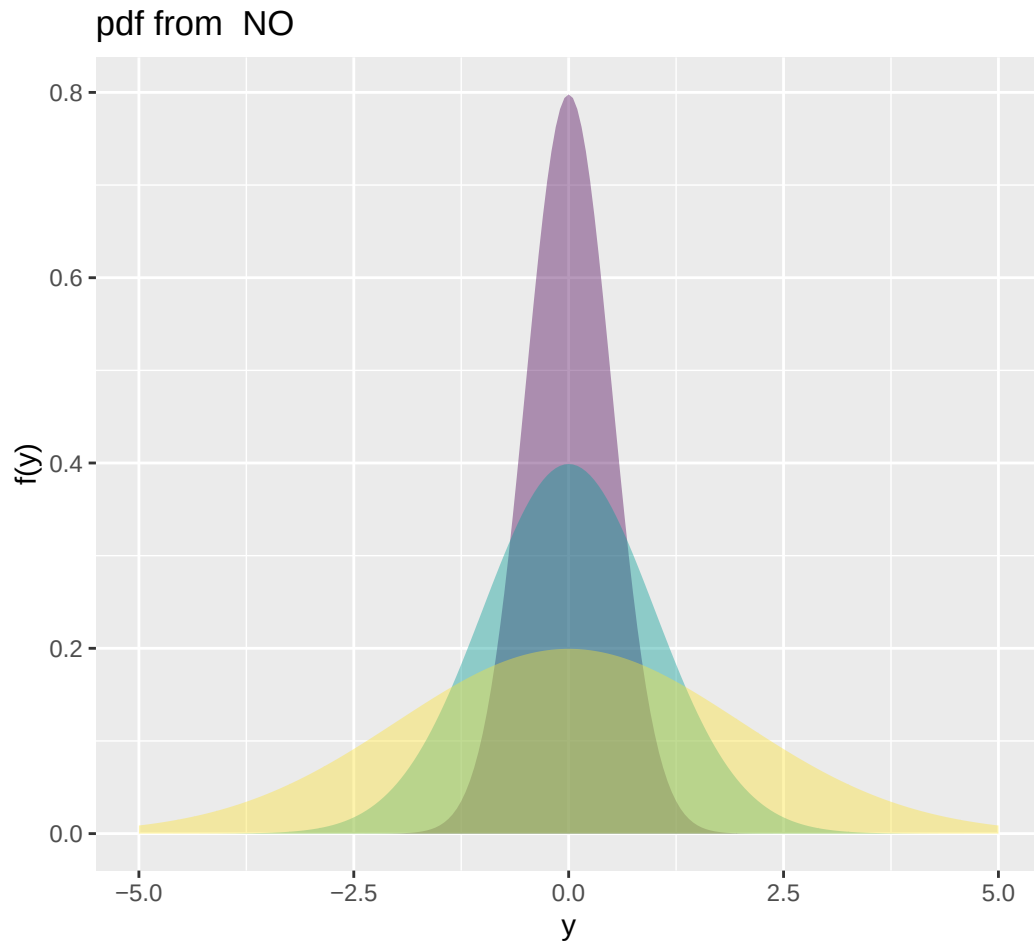


Figure 2: Continuous response example: plotting the pdf of a normal random variable.

The following demonstrates how discrete distributions are displayed. We begin with a distribution that can take infinitely many count values—the Negative Binomial distribution;

```
family_pdf(NBI, to=15, mu=1, sigma=c(.5,1,2), alpha=.9, size.segment = 3)
```

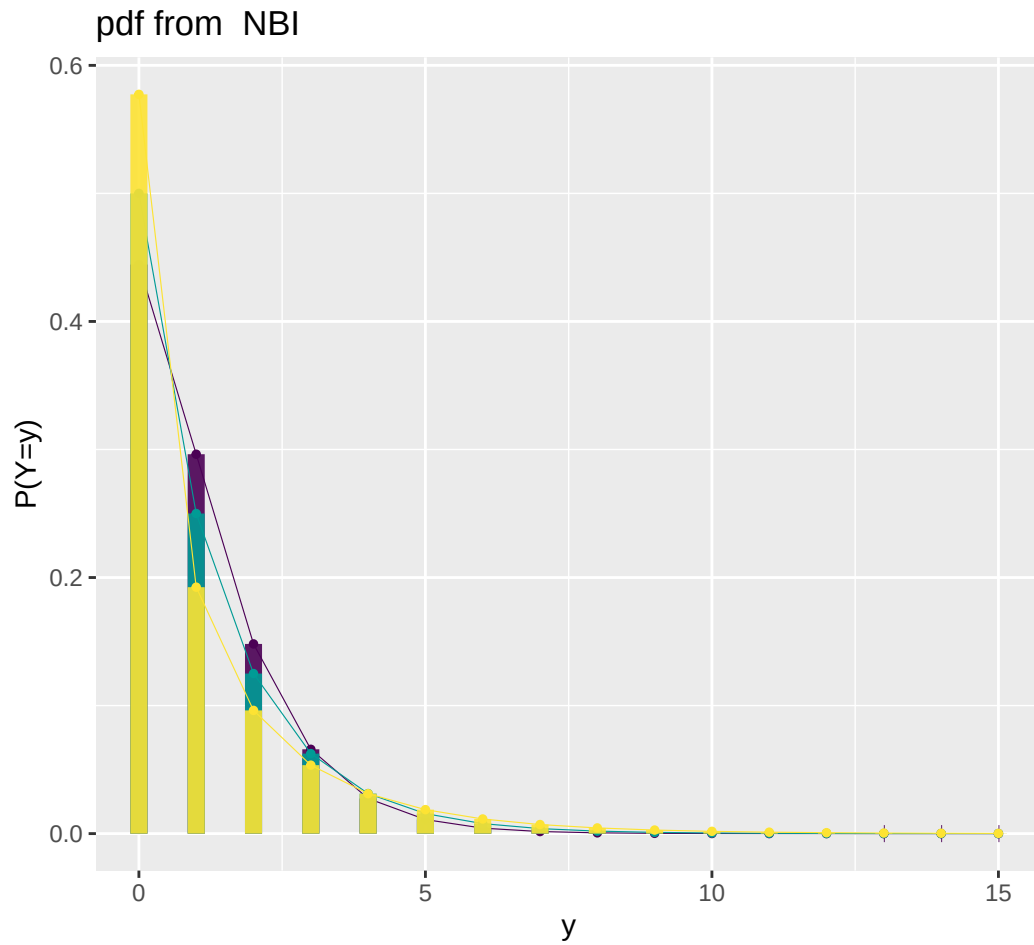


Figure 3: Count response example: plotting the pdf of a beta binomial.

Here we use a discrete distribution with a finite range of possible values: the Beta-Binomial distribution.

```
family_pdf(BB, to=15, mu=.5, sigma=c(.5,1,2), alpha=.9, , size.segment = 3)
```

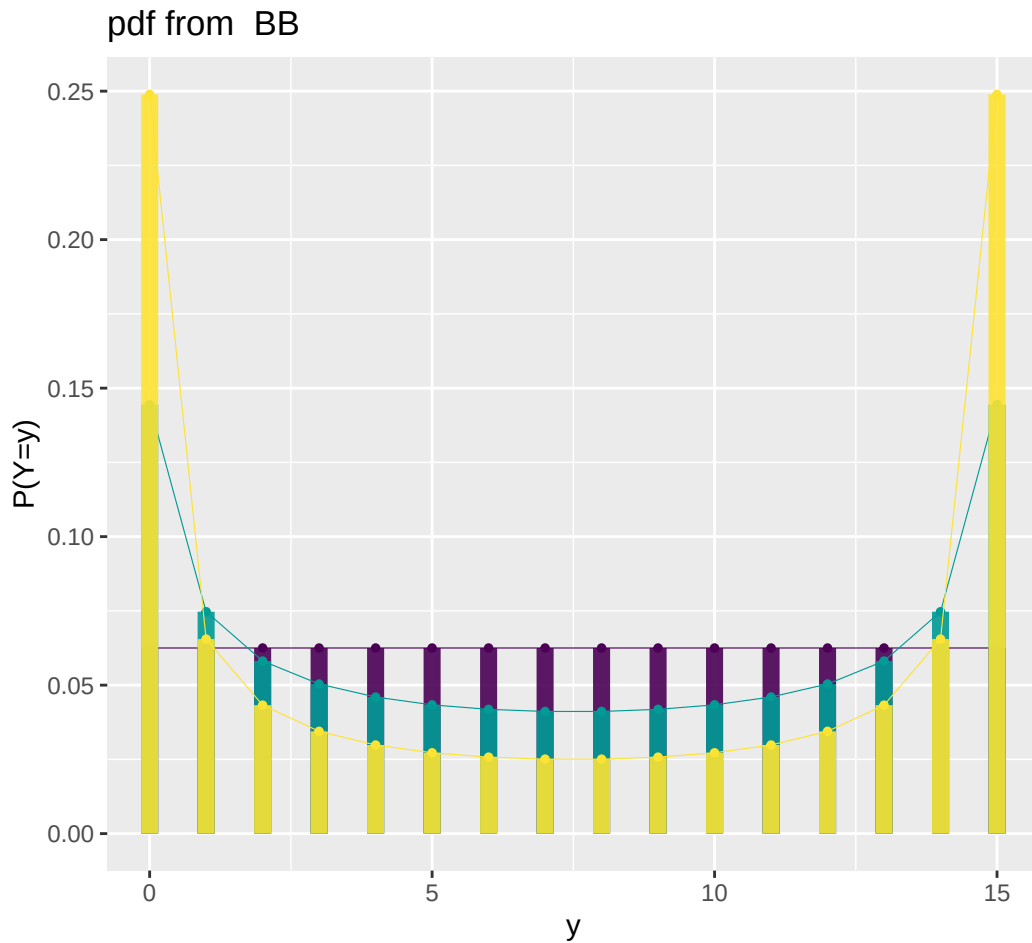


Figure 4: Beta binomial response example: plotting the pdf of a beta binomial.

`family_cdf`

### The function `family_cdf()`

The function `family_cdf()` plots cumulative distribution functions (CDFs) from the **`gamlss.family`** distributions. The primary argument required is the family specifying the distribution to be used.

```
family_cdf(NBI, to=15, mu=1, sigma=c(.5,1,2), alpha=.9, size.segment = 3)
```

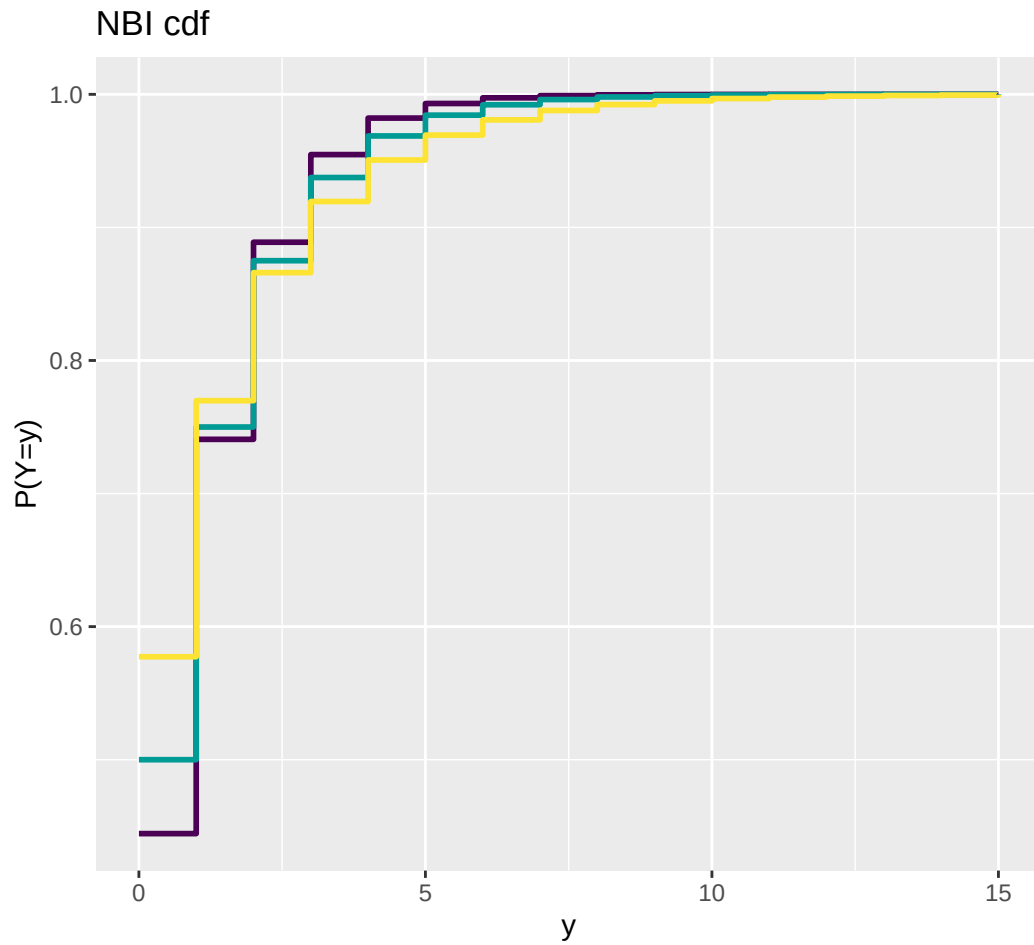


Figure 5: Count response example: plotting the cdf of a negative binomial.

The Cdf of the negative binomial;

```
family_cdf(BB, to=15, mu=.5, sigma=c(.5,1,2), alpha=.9, , size.segment = 3)
```

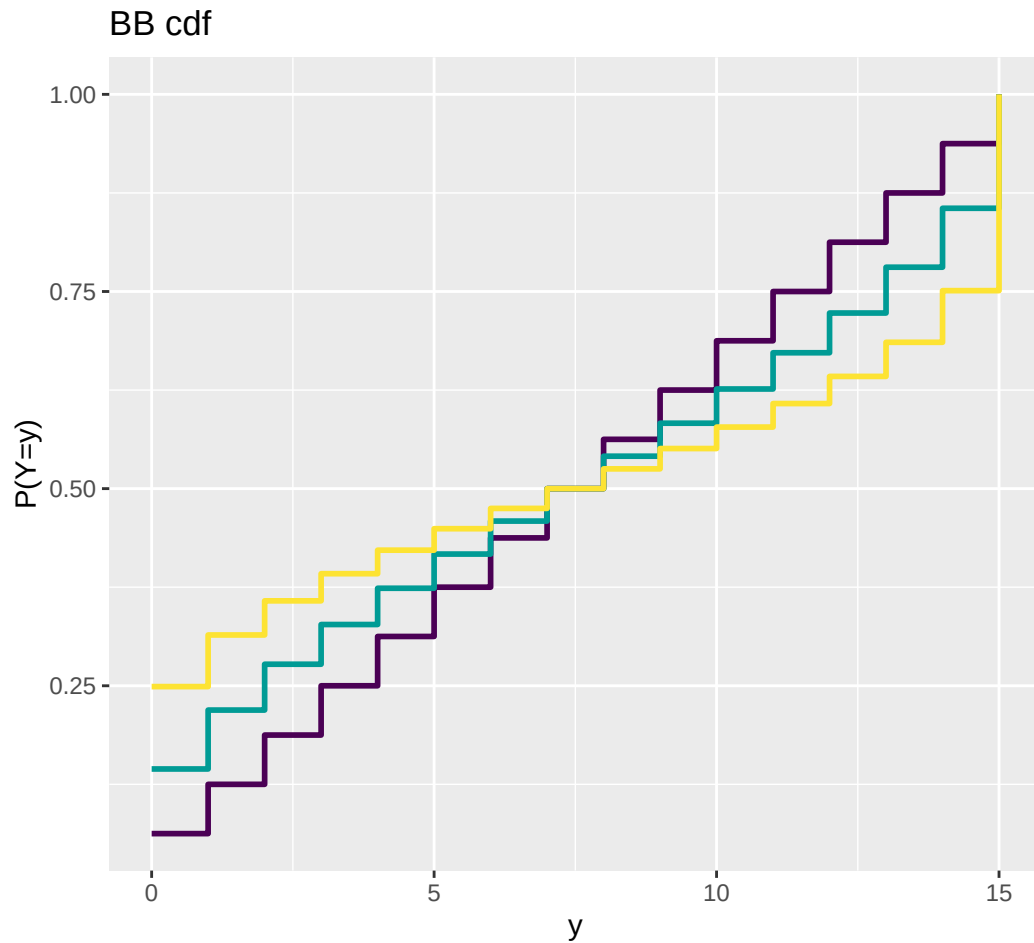


Figure 6: Count response example: plotting the cdf of a beta binomial.

#### `family_cor()`

The function `family_cor()` offers a basic method for examining the inter-correlation among the parameters of any distribution from the **gamlss.family**. It performs the following steps:

- Generates 10,000 random values from the specified distribution.
- Fits the same distribution to the generated data.
- Extracts and plots the correlation coefficients of the distributional parameters.

These correlation coefficients are derived from the variance-covariance matrix of the fitted model.

### Warning

This method provides only a rough indication of how the parameters are correlated at specific values of the distribution's parameters. The correlation structure may vary significantly at different points in the parameter space, as the distribution can behave quite differently depending on those values.

```
#source("~/Dropbox/github/gamlss-ggplots/R/family_cor.R")  
gamlss.ggplots::family_cor("BCTo", mu=1, sigma=0.11, nu=1, tau=5, no.sim=10000)
```

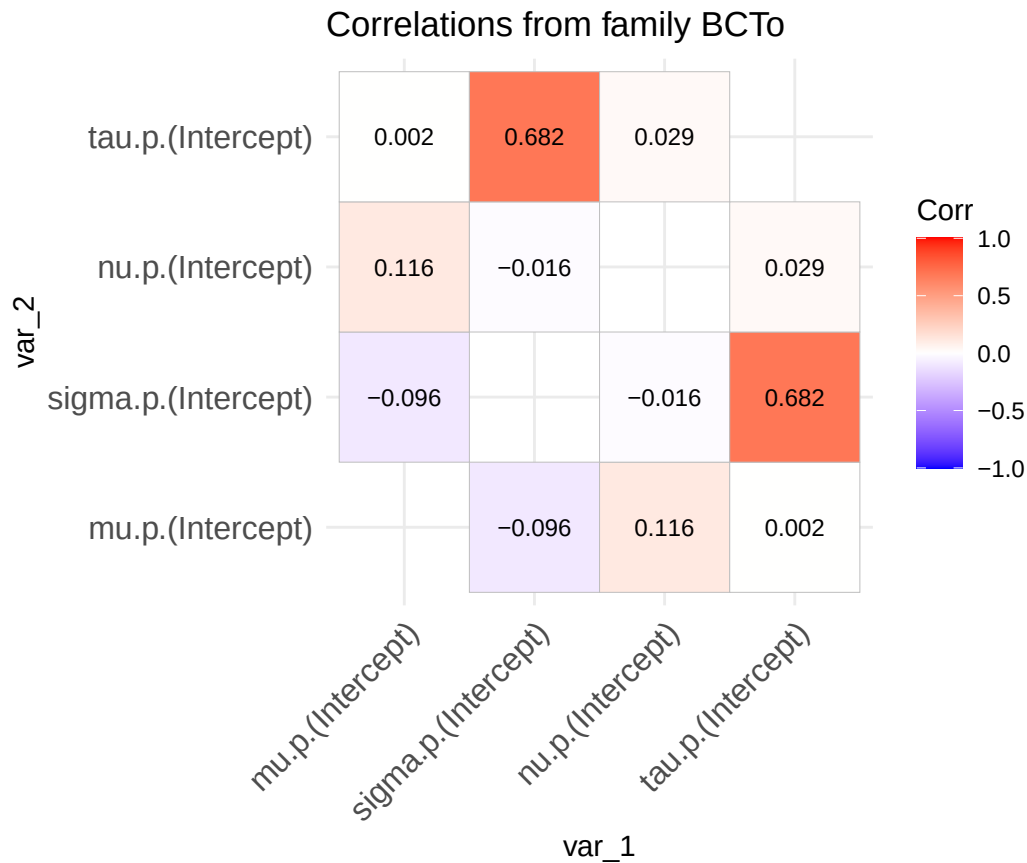


Figure 7: Family correlation of a BCTo distribution at specified values of the parameters.

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